భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

FLEXIBLE PRESSURE SENSORS FOR THERAPEUTIC ULTRASOUND APPLICATIONS

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Date: 11 july 2025

OBJECTIVES

- Synthesis of BCZT based pressure sensor.
- Structural and electrical characterisation of the sensor material.
- Integration and testing of the fabricated sensors with the therapeutic Ultrasound system.

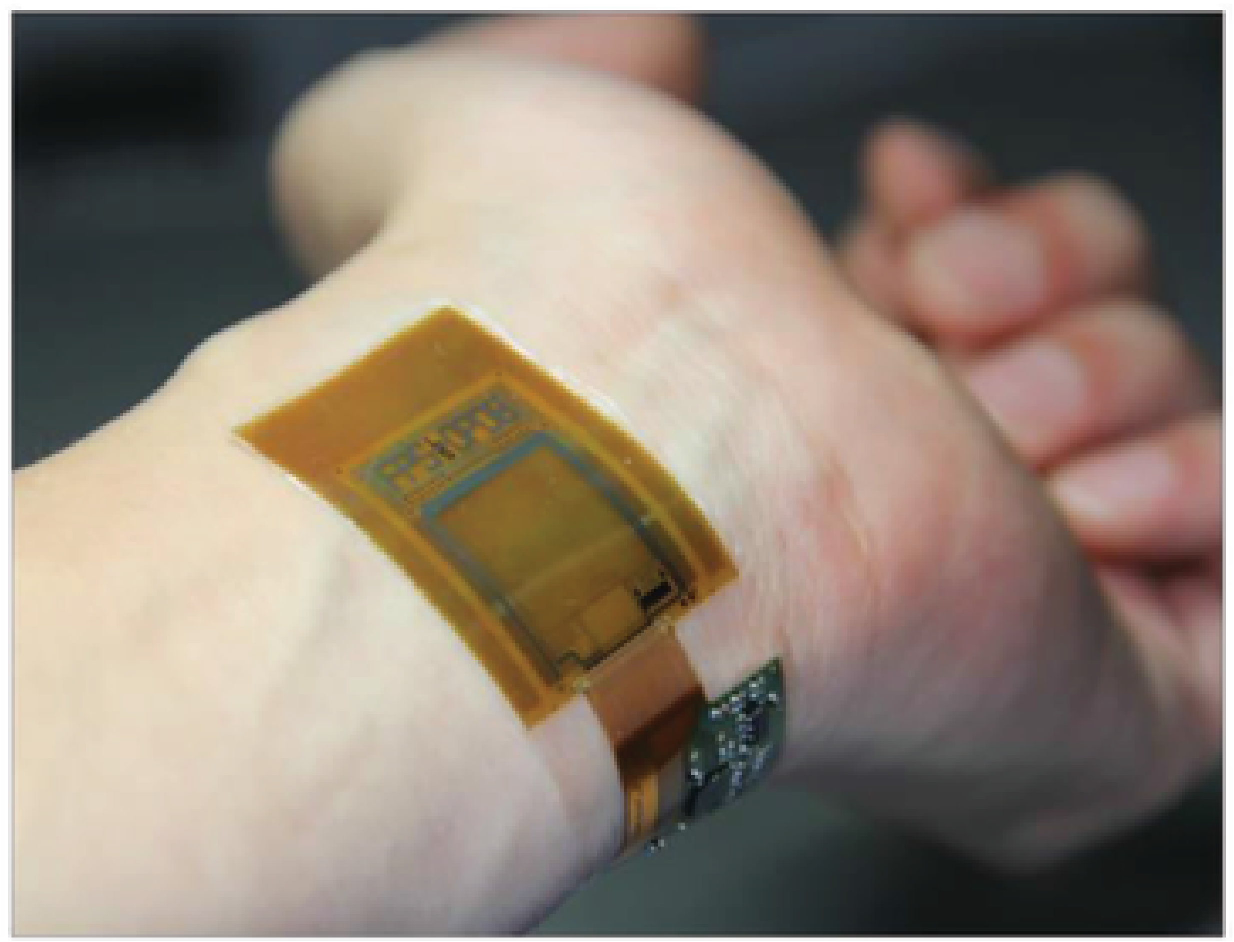


Fig1. Flexible sensors

Fabrication and testing of a flexible printed sensor prototype → Synthesis of piezoelectric composites and deposition on substrate → Characterization & calibration of the flexible sensors → Integration of the sensor with the therapeutic ultrasound system

Refrence: https://www.researchgate.net/figure/Flexible-image-sensor-with-high-resolution-and-high-speed-a-Picture-of-the-conformable_fig17_349006134 sensors,

1. Synthesis

- Powder synthesis
- Cleaning of powder prepared
- Annealing
- Mixing BCZT powder with PVDF polymer composites
- Spin coating over ITO coated PET
- Thermal evaporation

2. characterization & coding

IMPEDANCE ANALYZER

An impedance analyzer is an electronic instrument used to measure the electrical impedance (resistance to AC) of a material/device over a range of frequencies.

- **How Does It Work?**
  - The analyzer applies a small AC voltage across the sample.
  - Measures the resulting current response.
  - Calculates impedance (Z) as a function of frequency.

From this, various material parameters can be derived:

- **Capacitance (Cp)**
- **Dissipation Factor (D)**

impedance analyzer dataset format

FW: REV1.1

Meas. Parameter	Cp - D	Osc Level	0.5 [V]
Adapter	Probe, 42941A	DC Bias	Off
Sweep Type	LOG	BW	1
Number of Points	501	Sweep Averaging	16
Point Delay Time	0.1 [Sec]	Point Averaging	Off
Sweep Delay Time	0.1 [Sec]		

Graph Parameter	
X	Frequency
Y1	Cp
Y2	D

Point#	Frequency	Data Trace A Real	Data Trace A Imag	Data Trace B Real	Data Trace B Imag
1	1.0000E+02	-4.4847E-09	0.0000E+00	6.6666E+02	0.0000E+00
2	1.0233E+02	-1.9423E-09	0.0000E+00	1.5346E+03	0.0000E+00
3	1.0471E+02	-7.8689E-10	0.0000E+00	3.5232E+03	0.0000E+00
4	1.0715E+02	-5.2705E-09	0.0000E+00	5.1497E+02	0.0000E+00
5	1.0965E+02	-7.1411E-09	0.0000E+00	3.7081E+02	0.0000E+00
6	1.1220E+02	-8.7532E-10	0.0000E+00	2.9401E+03	0.0000E+00
7	1.1482E+02	-6.7143E-10	0.0000E+00	3.7543E+03	0.0000E+00
8	1.1749E+02	-2.2724E-09	0.0000E+00	1.0836E+03	0.0000E+00
9	1.2023E+02	-2.4161E-09	0.0000E+00	1.0065E+03	0.0000E+00
10	1.2303E+02	-4.7626E-11	0.0000E+00	5.3629E+04	0.0000E+00
11	1.2589E+02	-2.1384E-10	0.0000E+00	1.1662E+04	0.0000E+00
12	1.2882E+02	-8.6327E-10	0.0000E+00	2.8352E+03	0.0000E+00
13	1.3183E+02	-8.0634E-10	0.0000E+00	2.9541E+03	0.0000E+00
14	1.3490E+02	-7.2427E-10	0.0000E+00	3.2222E+03	0.0000E+00
15	1.3804E+02	-6.7982E-10	0.0000E+00	3.3564E+03	0.0000E+00
16	1.4125E+02	-8.0122E-10	0.0000E+00	2.7849E+03	0.0000E+00
17	1.4454E+02	-7.3044E-10	0.0000E+00	2.9855E+03	0.0000E+00
18	1.4791E+02	-6.2483E-10	0.0000E+00	3.4151E+03	0.0000E+00
19	1.5136E+02	-9.2641E-10	0.0000E+00	2.2545E+03	0.0000E+00

code for Cp vs f, D vs f & permittivity vs f

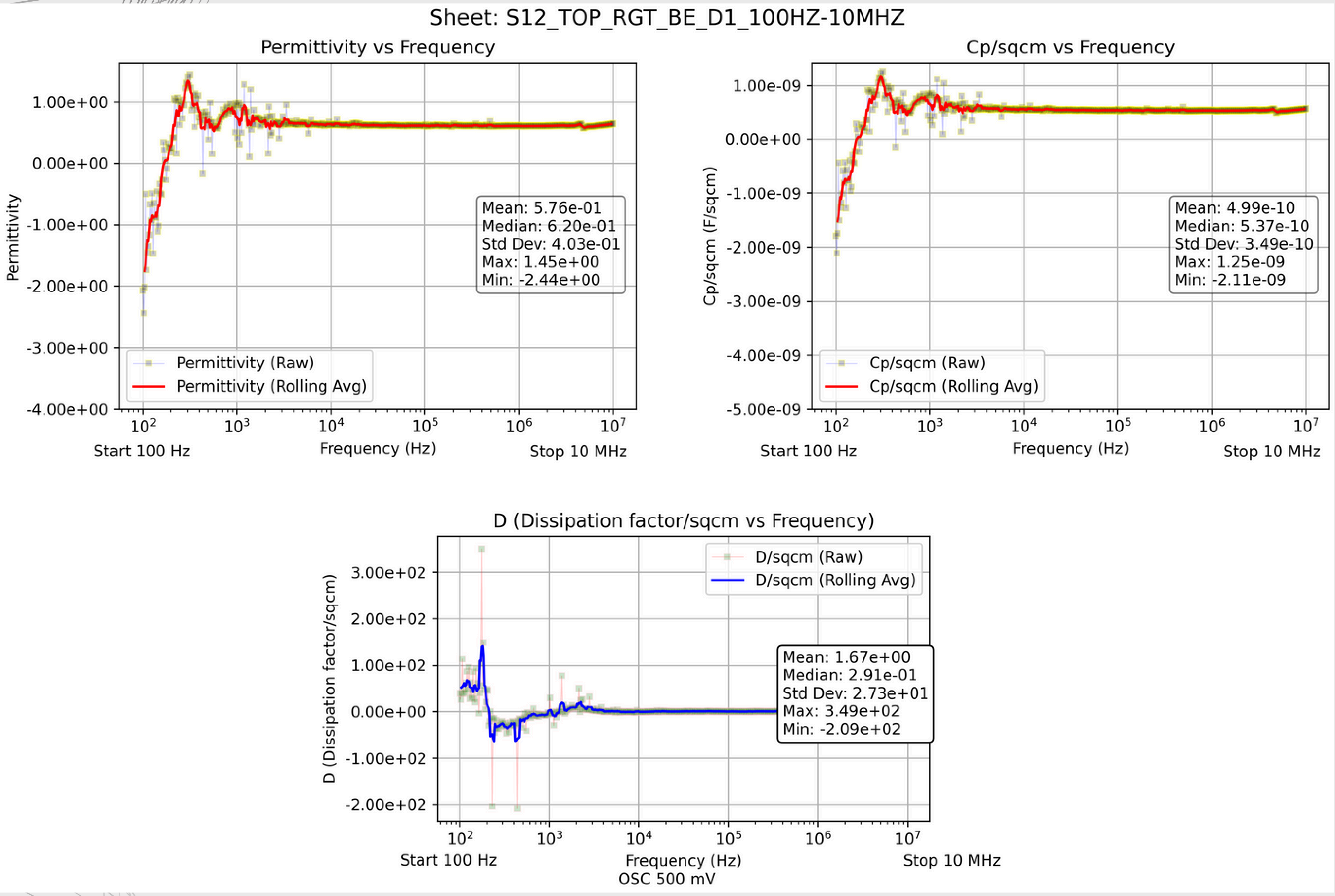
```python
x  = df['Frequency']
y3 = df['Data Trace A Real']                    # Cp
y4 = df['Data Trace B Real']                    # D
y1 = y3 / 0.03475                               # Cp / area
y2 = y4 / 0.03475                               # D / area
y1_avg = y1.rolling(window=5, center=True).mean()
y2_avg = y2.rolling(window=5, center=True).mean()

# choose thickness by group
if   'S11' in sheet_name: t, group = 1e-6,          'S11'
elif 'S12' in sheet_name: t, group = 1.02165e-6,    'S12'
elif 'S13' in sheet_name: t, group = 1.2315e-6,     'S13'
else:
    print(f"Warning: no thickness for {sheet_name}; skipping")
    continue

ye      = y3 * t / (8.854187817e-12 * 3.475e-6)       # permittivity
ye_avg  = ye.rolling(window=5, center=True).mean()
```

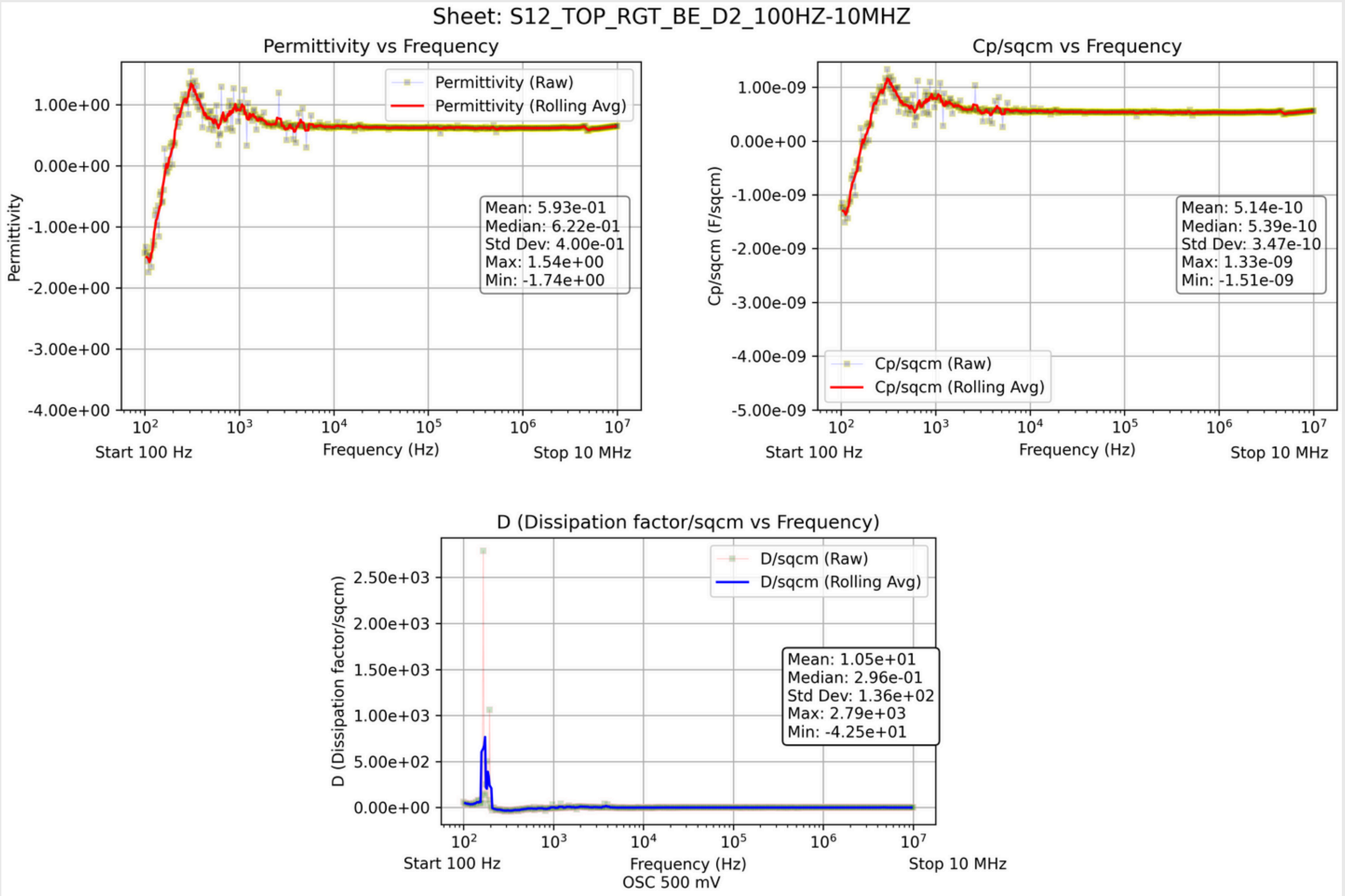
$$C = \frac{\varepsilon_0\,\varepsilon_r\,A}{d}$$

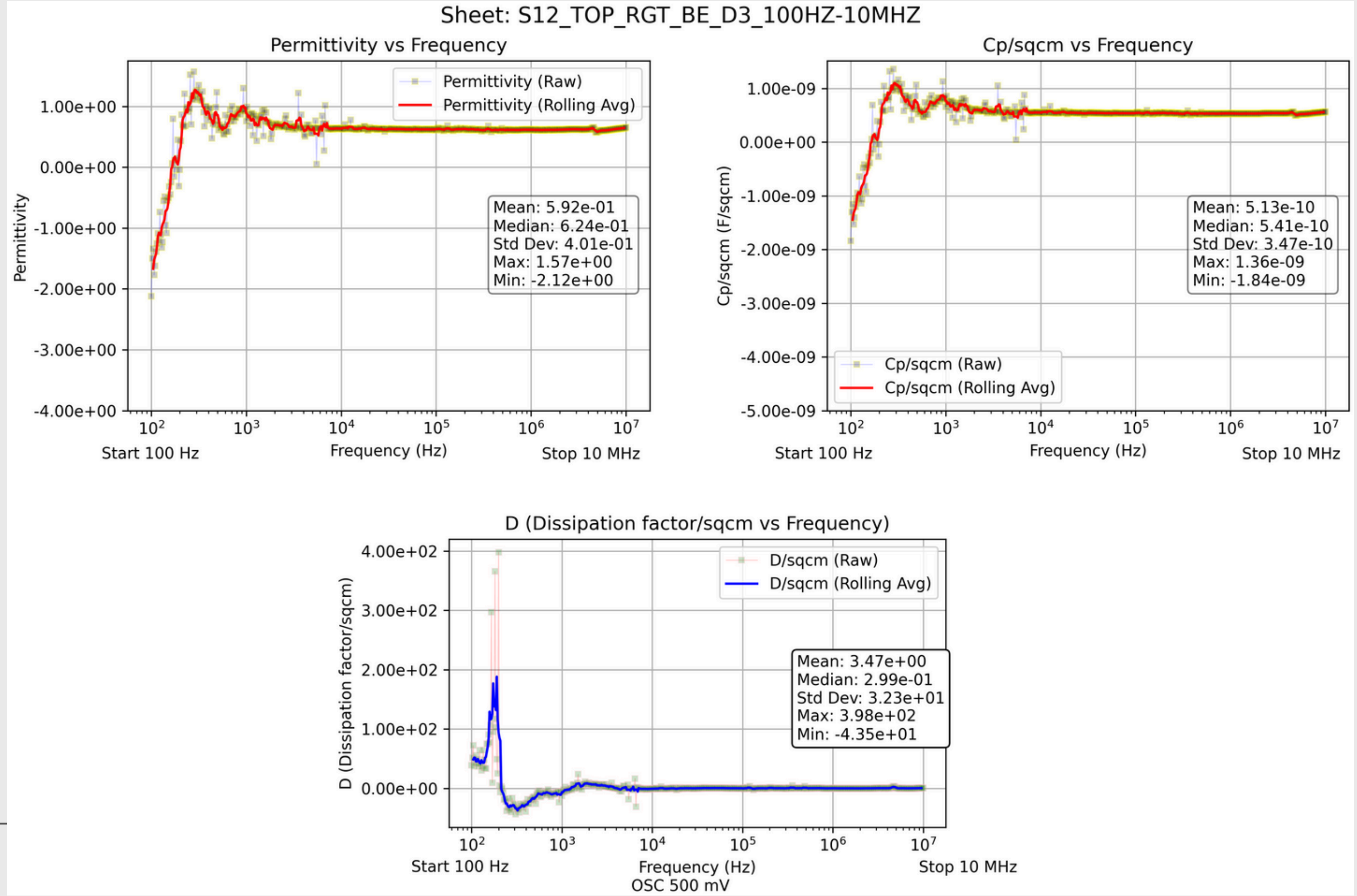
$$\varepsilon_r = \frac{C \cdot d}{\varepsilon_0 \cdot A}$$

Impedance Code: https://drive.google.com/file/d/1hQ9JvMc_etM5htEGxfu8d3TC_KGN7qgk/view?usp=sharing

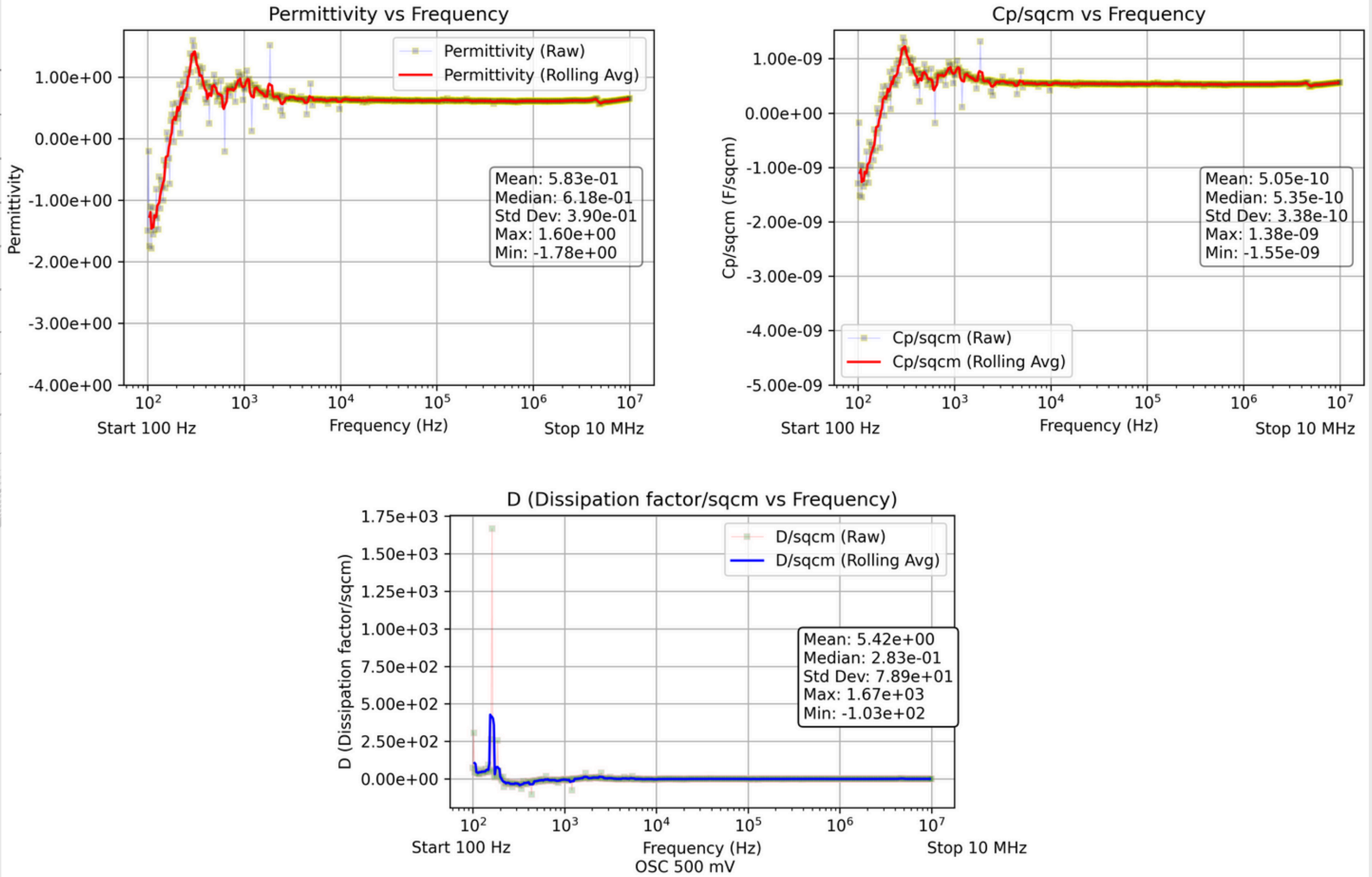
output: Sample_S12



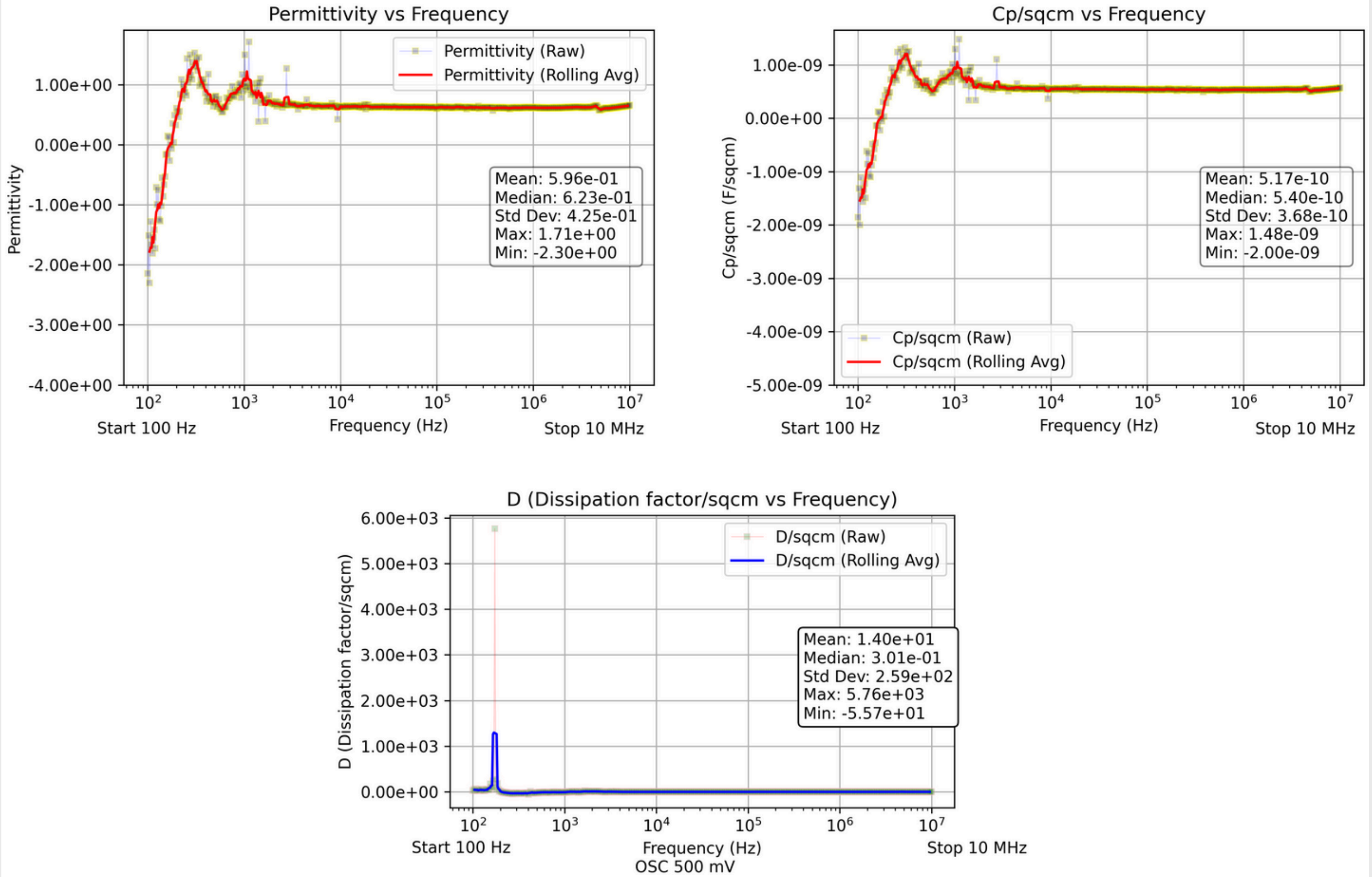




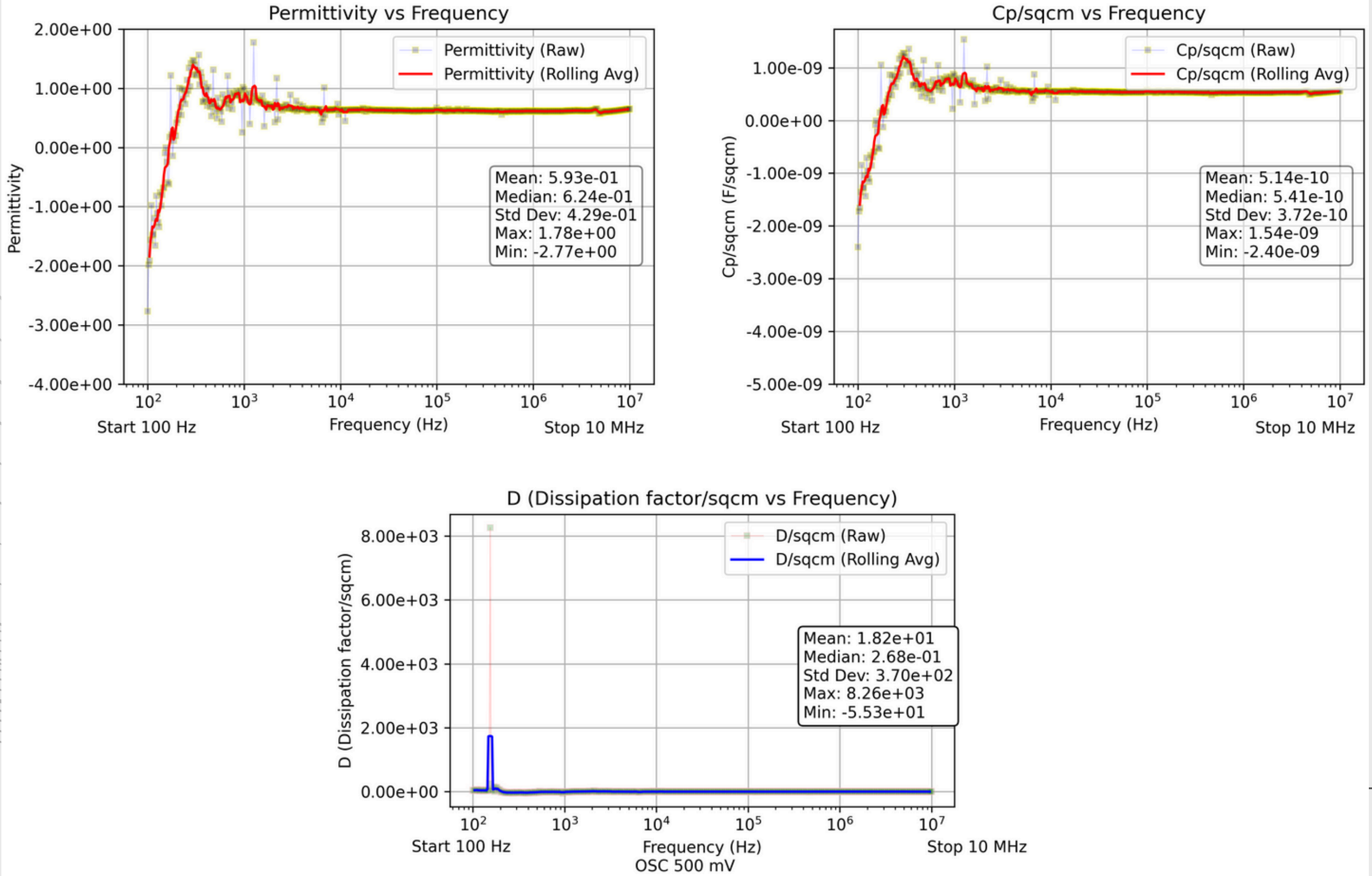
Sheet: S12_TOP_RGT_BE_D4_100HZ-10MHZ



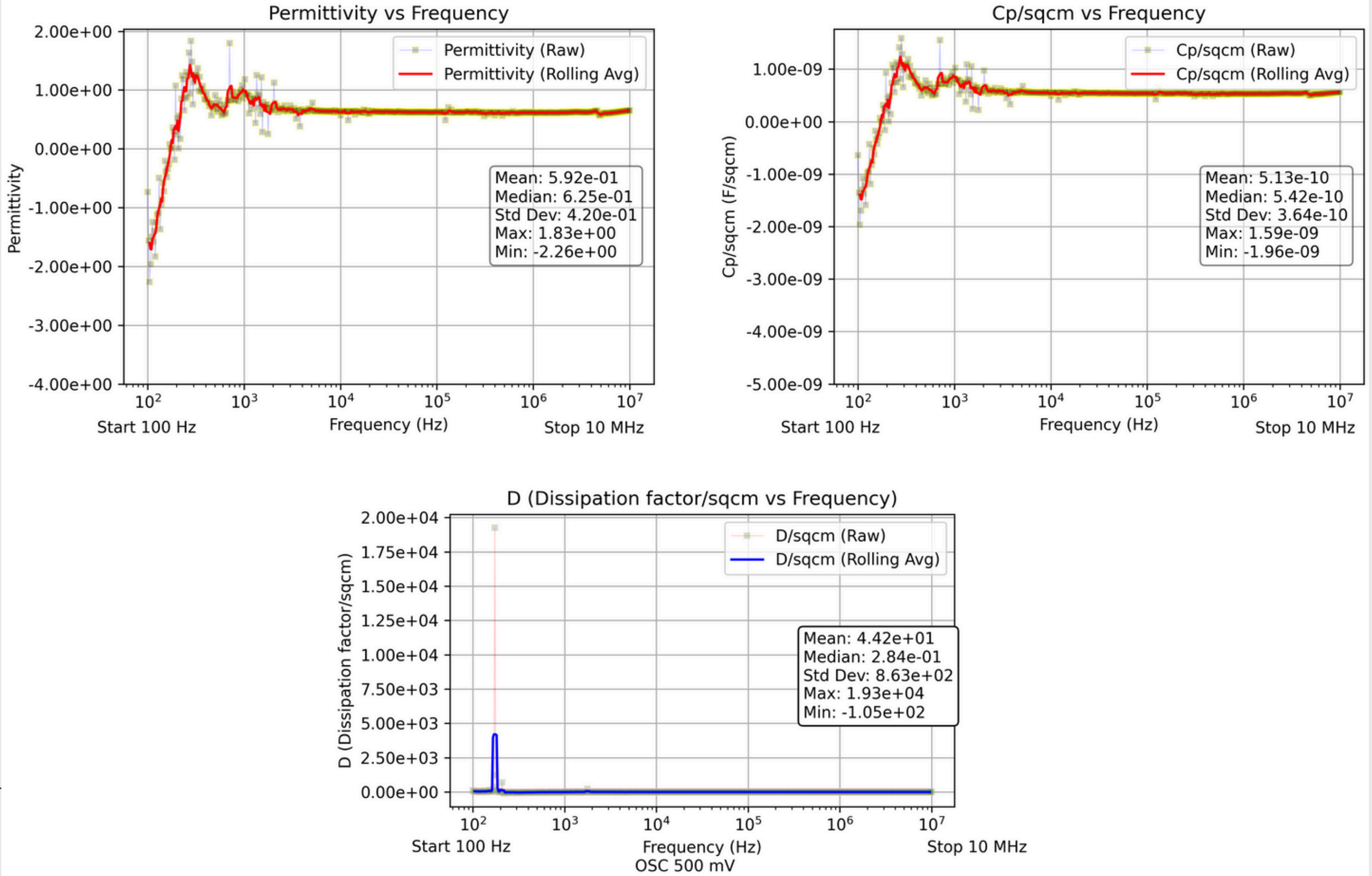
Sheet: S12_TOP_RGT_BE_D5_100HZ-10MHZ



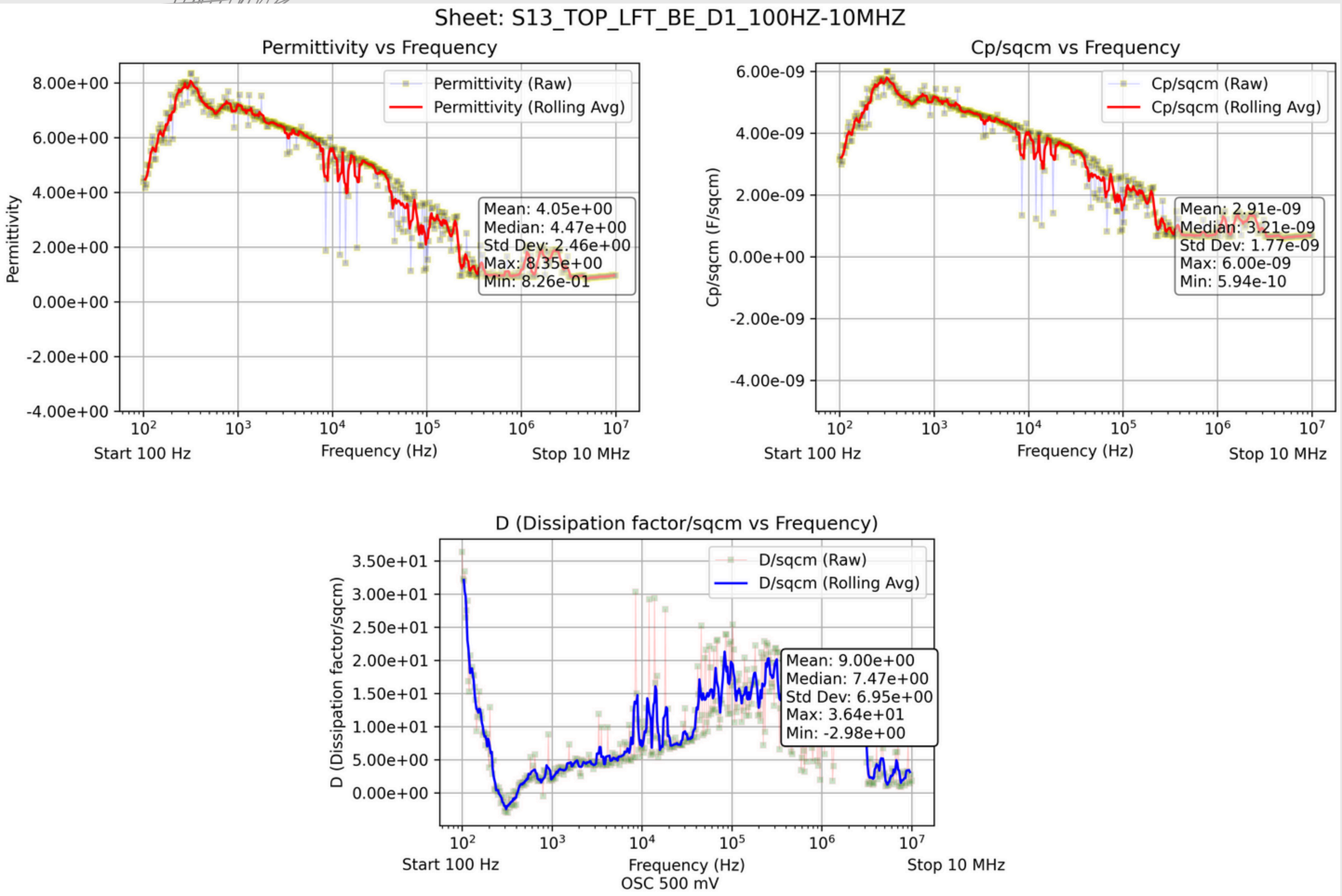
Sheet: S12_TOP_RGT_BE_D6_100HZ-10MHZ

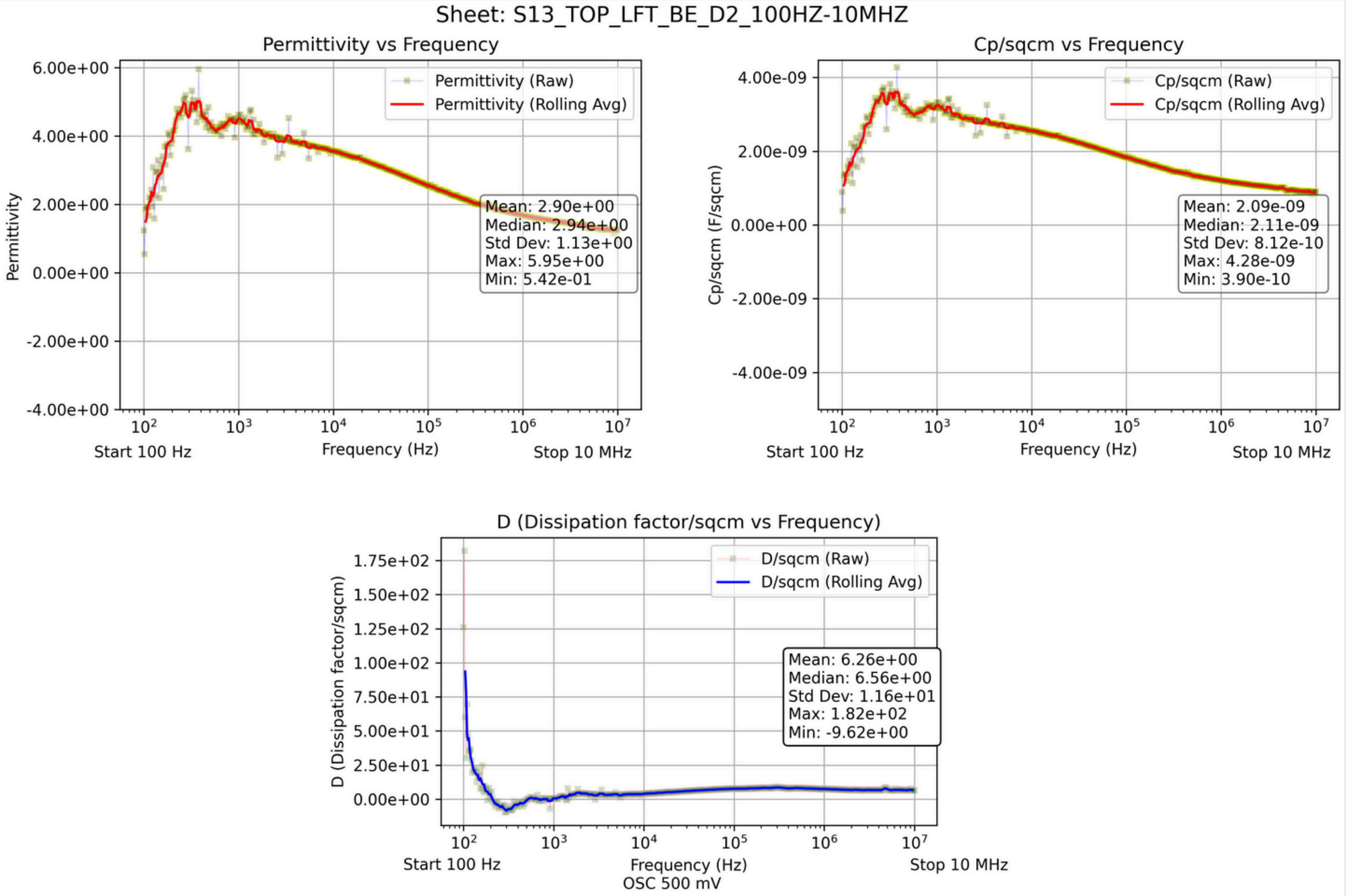


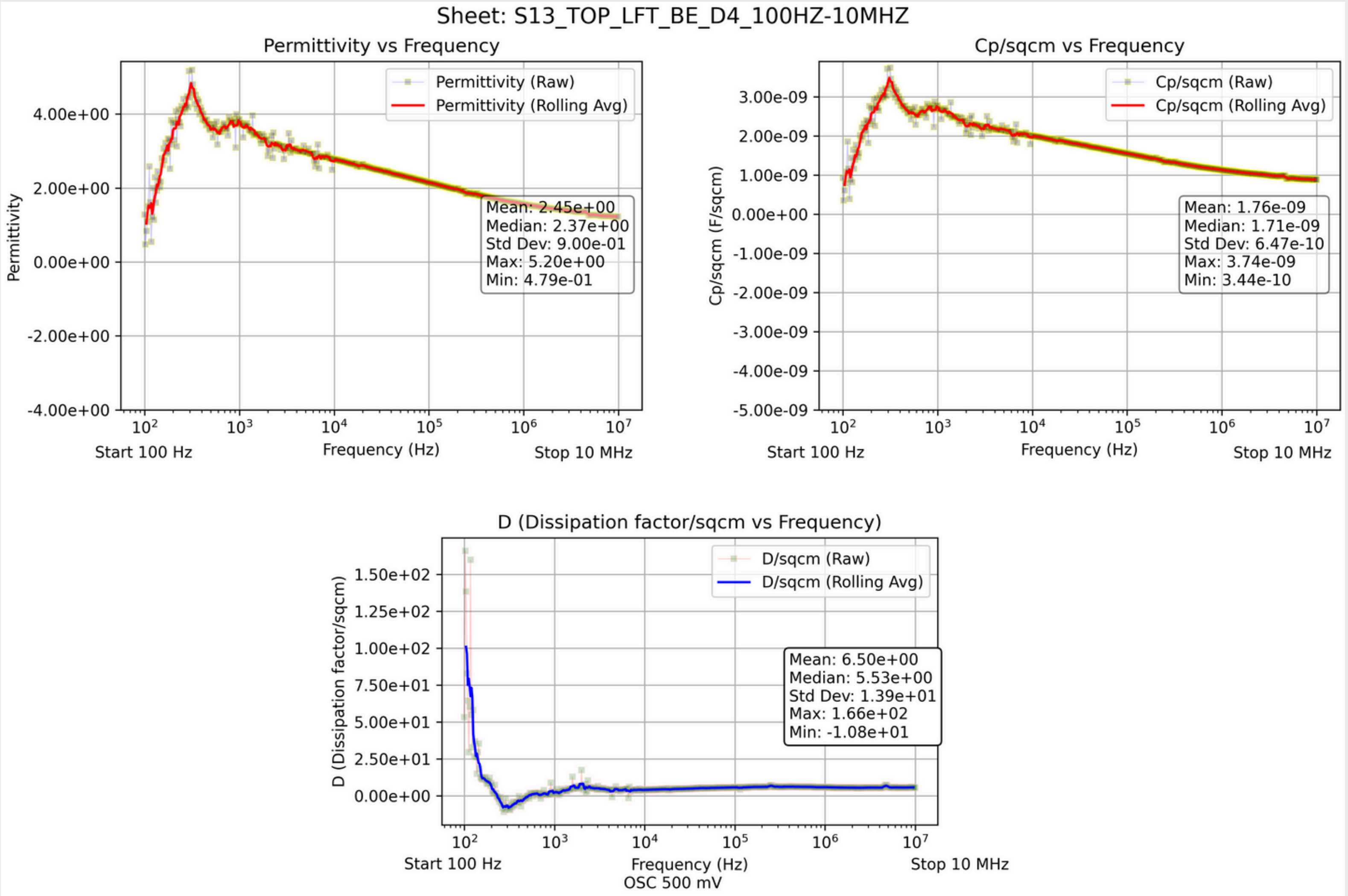
Sheet: S12_TOP_RGT_BE_D7_100HZ-10MHZ

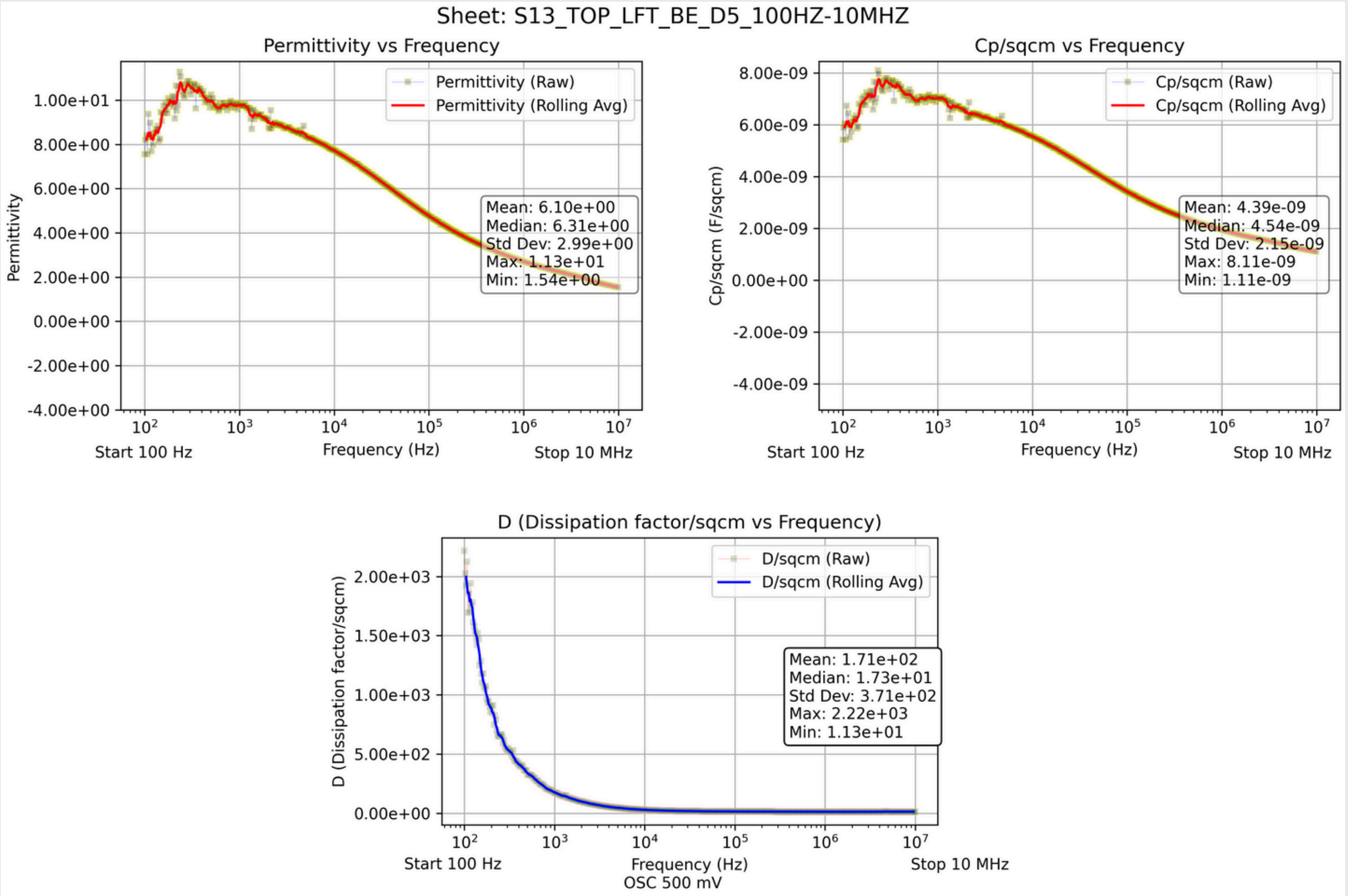


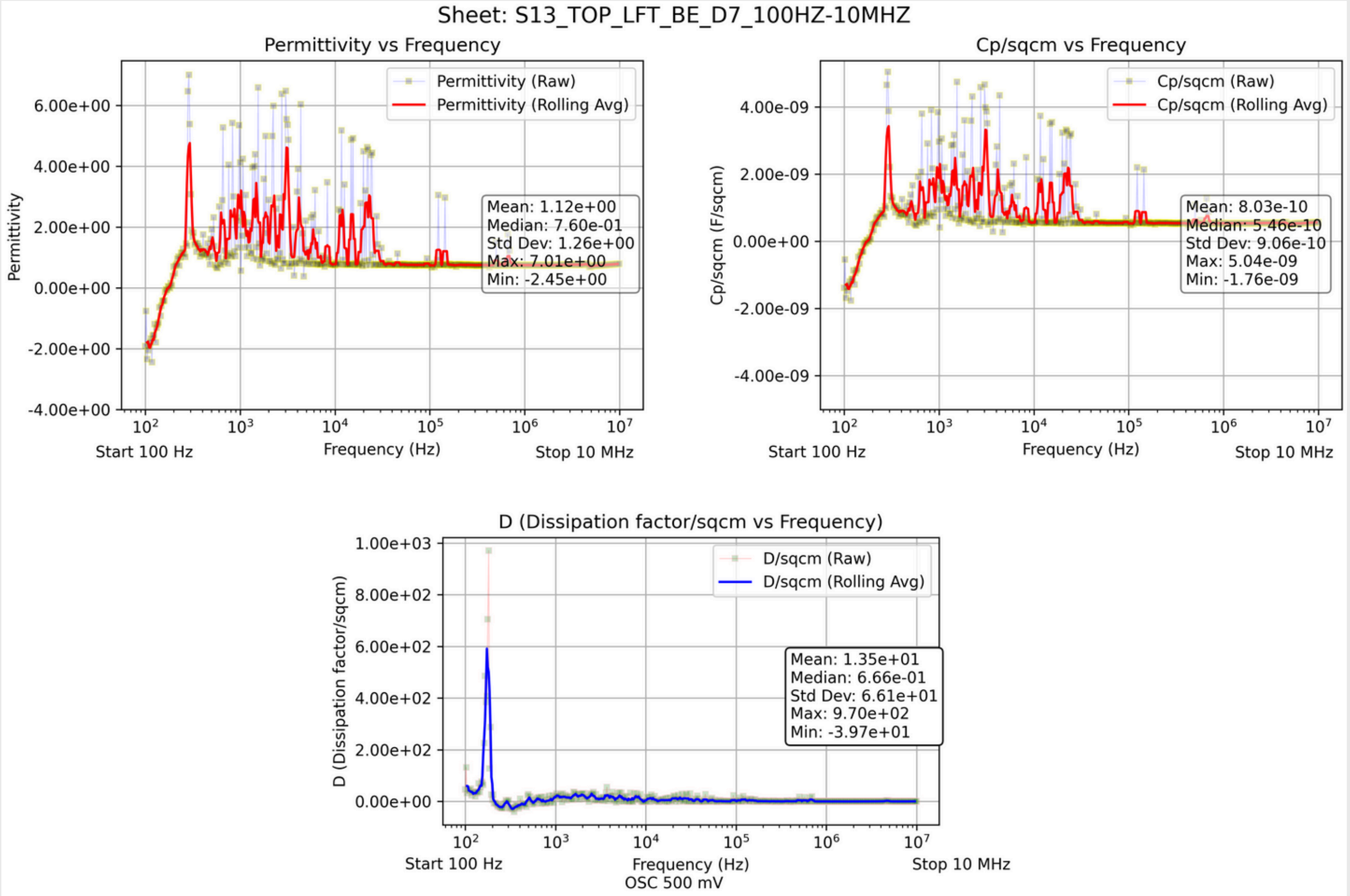
output: Sample S13






Sheet: S13_TOP_LFT_BE_D4_100HZ-10MHZ
Permittivity vs Frequency
Permittivity (Raw)
Permittivity (Rolling Avg)
Mean: 2.45e+00
Median: 2.37e+00
Std Dev: 9.00e-01
Max: 5.20e+00
Min: 4.79e-01
Permittivity
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
Cp/sqcm vs Frequency
Cp/sqcm (Raw)
Cp/sqcm (Rolling Avg)
Mean: 1.76e-09
Median: 1.71e-09
Std Dev: 6.47e-10
Max: 3.74e-09
Min: 3.44e-10
Cp/sqcm (F/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
D (Dissipation factor/sqcm vs Frequency)
D/sqcm (Raw)
D/sqcm (Rolling Avg)
Mean: 6.50e+00
Median: 5.53e+00
Std Dev: 1.39e+01
Max: 1.66e+02
Min: -1.08e+01
D (Dissipation factor/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
OSC 500 mV


Sheet: S13_TOP_LFT_BE_D5_100HZ-10MHZ
Permittivity vs Frequency
Permittivity (Raw)
Permittivity (Rolling Avg)
Mean: 6.10e+00
Median: 6.31e+00
Std Dev: 2.99e+00
Max: 1.13e+01
Min: 1.54e+00
Permittivity
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
Cp/sqcm vs Frequency
Cp/sqcm (Raw)
Cp/sqcm (Rolling Avg)
Mean: 4.39e-09
Median: 4.54e-09
Std Dev: 2.15e-09
Max: 8.11e-09
Min: 1.11e-09
Cp/sqcm (F/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
D (Dissipation factor/sqcm vs Frequency)
D/sqcm (Raw)
D/sqcm (Rolling Avg)
Mean: 1.71e+02
Median: 1.73e+01
Std Dev: 3.71e+02
Max: 2.22e+03
Min: 1.13e+01
D (Dissipation factor/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
OSC 500 mV


Sheet: S13_TOP_LFT_BE_D7_100HZ-10MHZ
Permittivity vs Frequency
Permittivity (Raw)
Permittivity (Rolling Avg)
Mean: 1.12e+00
Median: 7.60e-01
Std Dev: 1.26e+00
Max: 7.01e+00
Min: -2.45e+00
Permittivity
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
Cp/sqcm vs Frequency
Cp/sqcm (Raw)
Cp/sqcm (Rolling Avg)
Mean: 8.03e-10
Median: 5.46e-10
Std Dev: 9.06e-10
Max: 5.04e-09
Min: -1.76e-09
Cp/sqcm (F/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
D (Dissipation factor/sqcm vs Frequency)
D/sqcm (Raw)
D/sqcm (Rolling Avg)
Mean: 1.35e+01
Median: 6.66e-01
Std Dev: 6.61e+01
Max: 9.70e+02
Min: -3.97e+01
D (Dissipation factor/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
OSC 500 mV

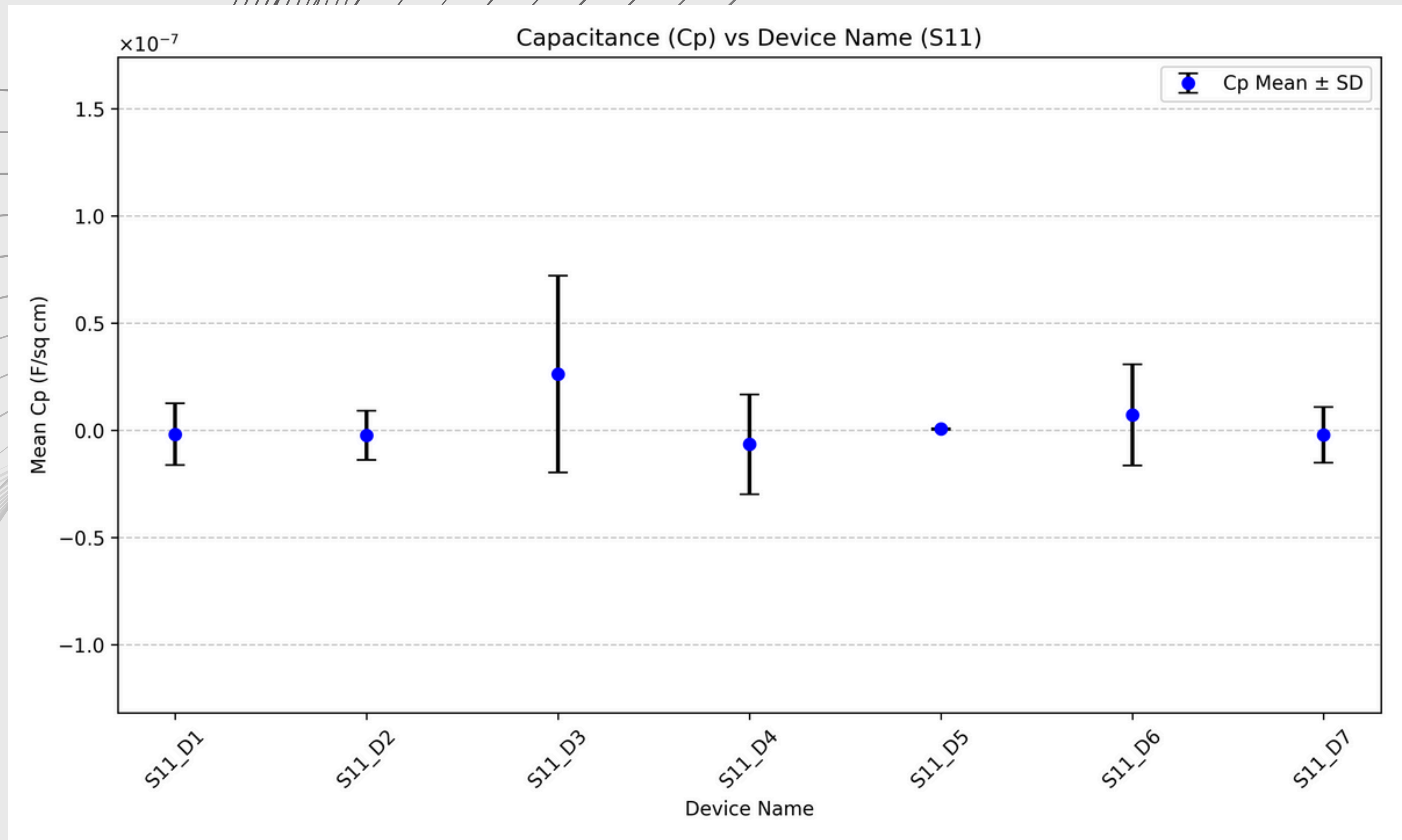
Summary Table

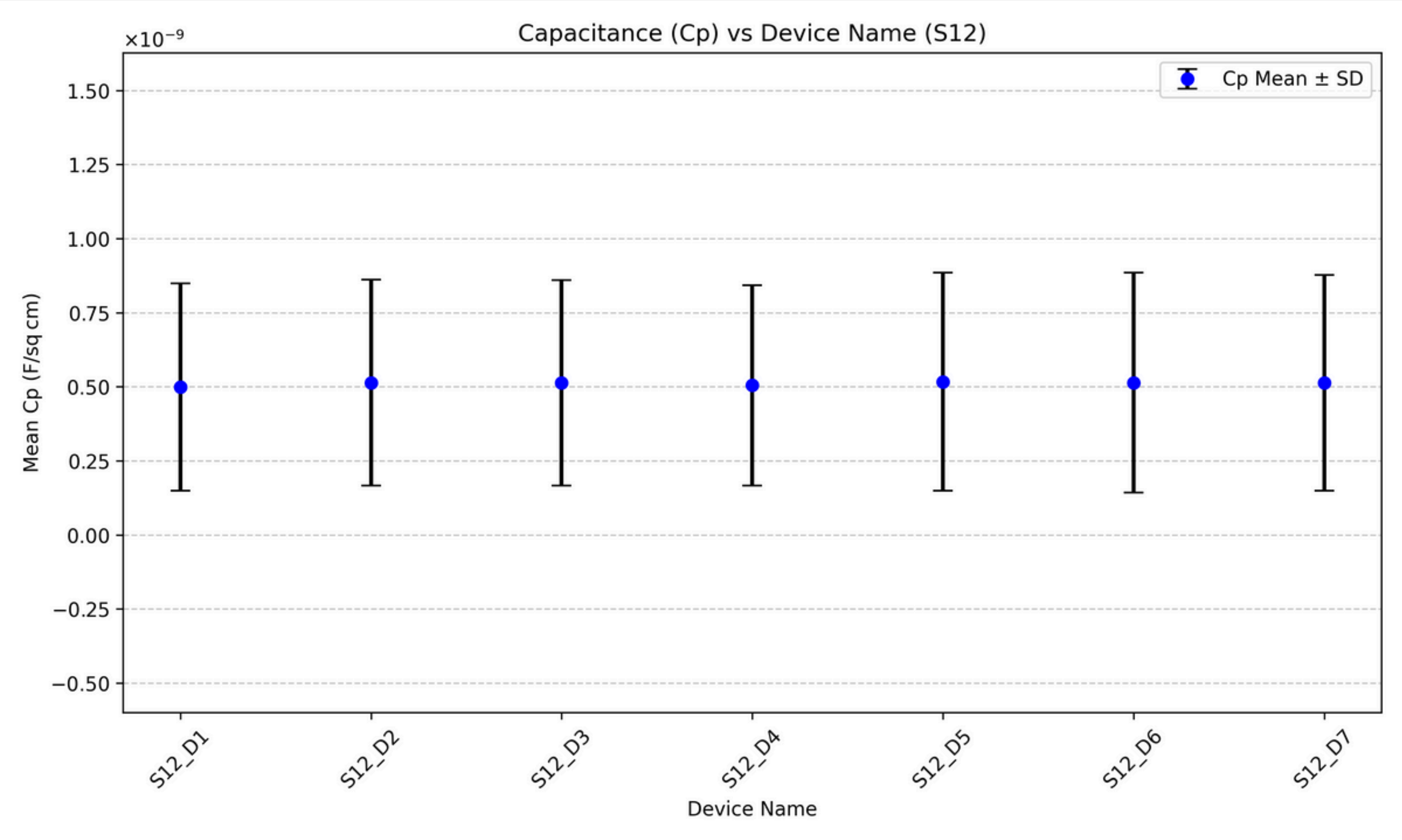
Group	Device Name	Mean Permittivity	Mean Cp	Mean Dissipation
S11	S11_D1	-2.12e+00±1.63e+01	-1.88e-09±1.44e-08	1.03e+05±3.23e+05
	S11_D2	-2.74e+00±1.30e+01	-2.43e-09±1.15e-08	1.24e+05±3.98e+05
	S11_D3	2.95e+01±5.18e+01	2.62e-08±4.59e-08	5.46e+03±5.54e+03
	S11_D4	-7.44e+00±2.63e+01	-6.59e-09±2.33e-08	1.25e+05±3.34e+05
	S11_D5	5.72e-01±4.06e-01	5.06e-10±3.60e-10	3.68e+01±1.60e+02
	S11_D6	7.95e+00±2.67e+01	7.04e-09±2.36e-08	1.25e+05±7.64e+05
	S11_D7	-2.41e+00±1.47e+01	-2.13e-09±1.30e-08	1.21e+05±3.46e+05
S12	S12_D1	5.76e-01±4.03e-01	4.99e-10±3.49e-10	1.67e+00±2.73e+01
	S12_D2	5.93e-01±4.00e-01	5.14e-10±3.47e-10	1.05e+01±1.36e+02
	S12_D3	5.92e-01±4.00e-01	5.13e-10±3.47e-10	3.47e+00±3.23e+01
	S12_D4	5.83e-01±3.89e-01	5.05e-10±3.38e-10	5.42e+00±7.89e+01
	S12_D5	5.96e-01±4.25e-01	5.17e-10±3.68e-10	1.40e+01±2.58e+02
	S12_D6	5.93e-01±4.28e-01	5.14e-10±3.71e-10	1.82e+01±3.69e+02
	S12_D7	5.92e-01±4.20e-01	5.13e-10±3.64e-10	4.42e+01±8.62e+02
S13	S13_D1	4.05e+00±2.46e+00	2.91e-09±1.77e-09	9.00e+00±6.94e+00
	S13_D2	2.90e+00±1.13e+00	2.09e-09±8.11e-10	6.26e+00±1.16e+01
	S13_D4	2.45e+00±8.99e-01	1.76e-09±6.47e-10	6.50e+00±1.39e+01
	S13_D5	6.10e+00±2.99e+00	4.39e-09±2.15e-09	1.71e+02±3.71e+02
	S13_D7	1.12e+00±1.26e+00	8.03e-10±9.05e-10	1.35e+01±6.60e+01

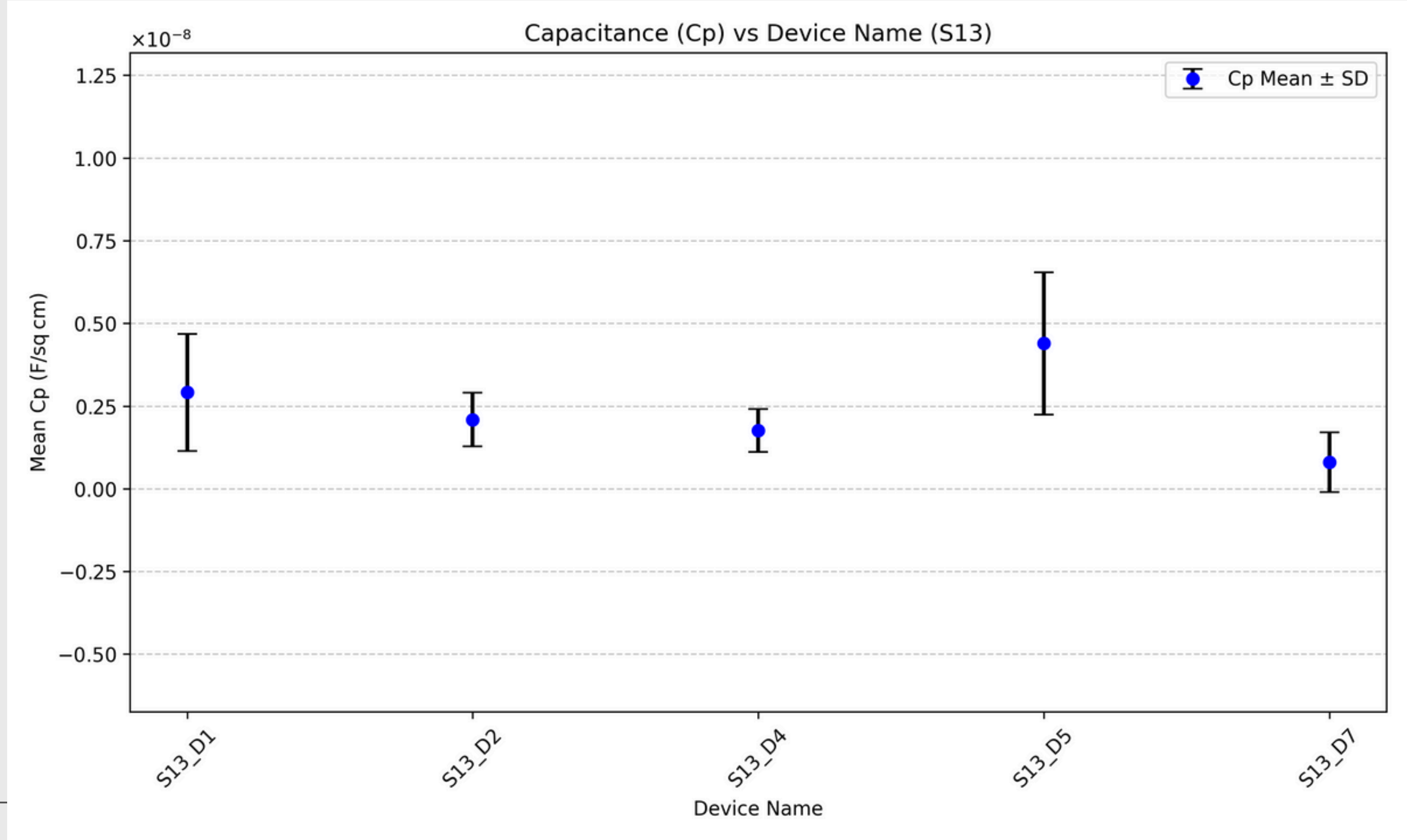
code for Cp vs Device & Permittivity vs Device

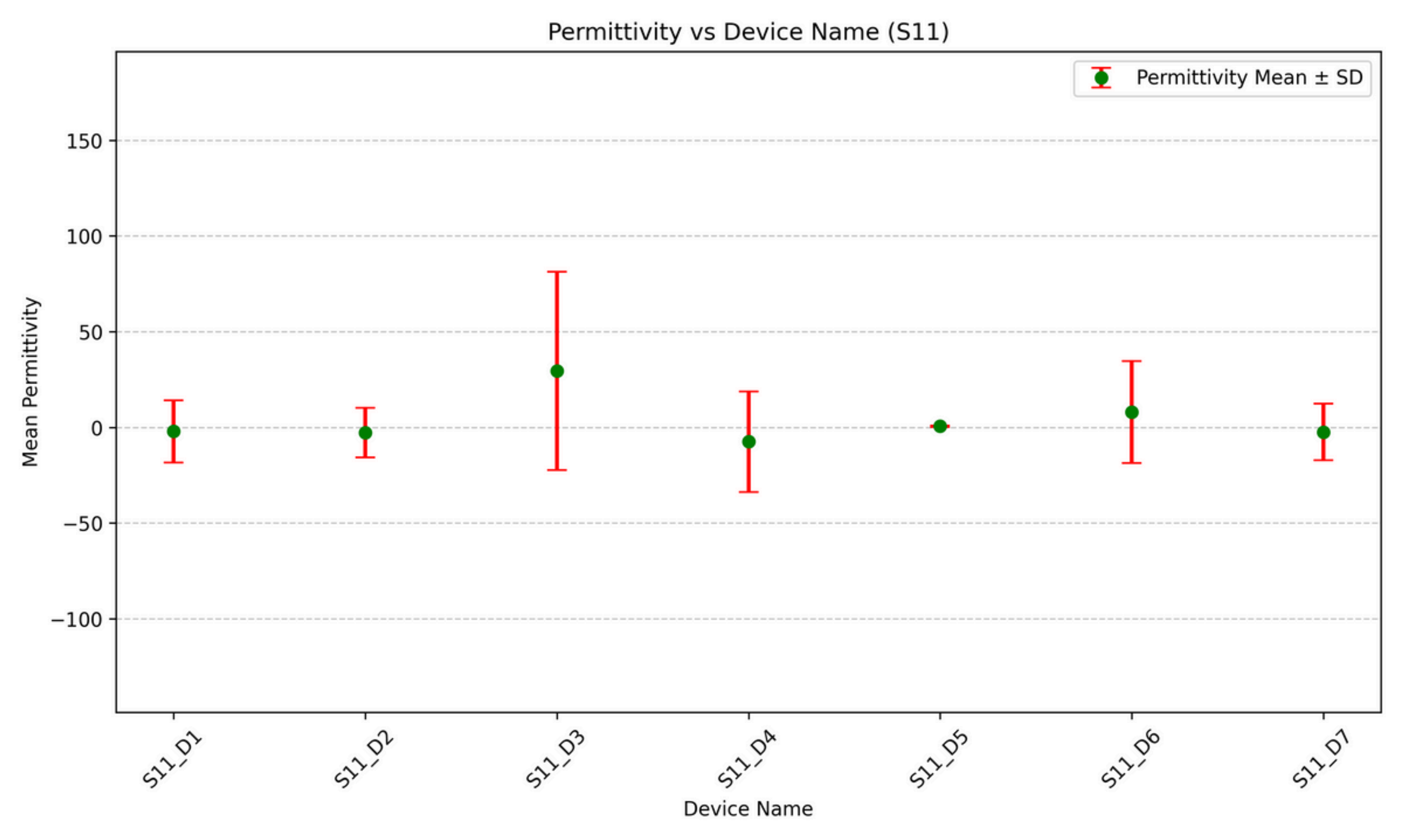
```
groups = ['S11', 'S12', 'S13']

for group in groups:
    gdf = df[df['Group'] == group].reset_index(drop=True)

    # ================== Permittivity plot ==================
    fig, ax = plt.subplots(figsize=(10, 6))
    ax.errorbar(gdf['Device Name'], gdf['Perm_Mean'],
                yerr=gdf['Perm_SD'], fmt='o',
                color='green', ecolor='red',
                elinewidth=2, capsize=5,
                label='Permittivity Mean ± SD')

    ax.set_ylim(*limits(gdf, 'Perm_Mean', 'Perm_SD'))
    ax.set_title(f'Permittivity vs Device Name ({group})')
    ax.set_xlabel('Device Name')
    ax.set_ylabel('Mean Permittivity')
    ax.ticklabel_format(axis='y', style='sci', scilimits=(-2, 2))
    ax.yaxis.set_major_formatter(ScalarFormatter(useMathText=True))
    plt.xticks(rotation=45)
    ax.grid(True, axis='y', linestyle='--', alpha=0.7)
    ax.legend()

    # move_offset(ax, x=-0.2, y=1)          # <-- shift
    plt.tight_layout()
    # save figure
    plots_dir = r"D:\recovery\error"
    os.makedirs(plots_dir, exist_ok=True)
    plt.savefig(os.path.join(plots_dir, f"{group}_Permittivity_plot.png"),dpi=300)
    # plt.show()
```

Cp/Permittivity vs Device: https://drive.google.com/file/d/1-jY2tYogKXxJQ9IapWUYvuH0Yd9AgFng/view?usp=sharing

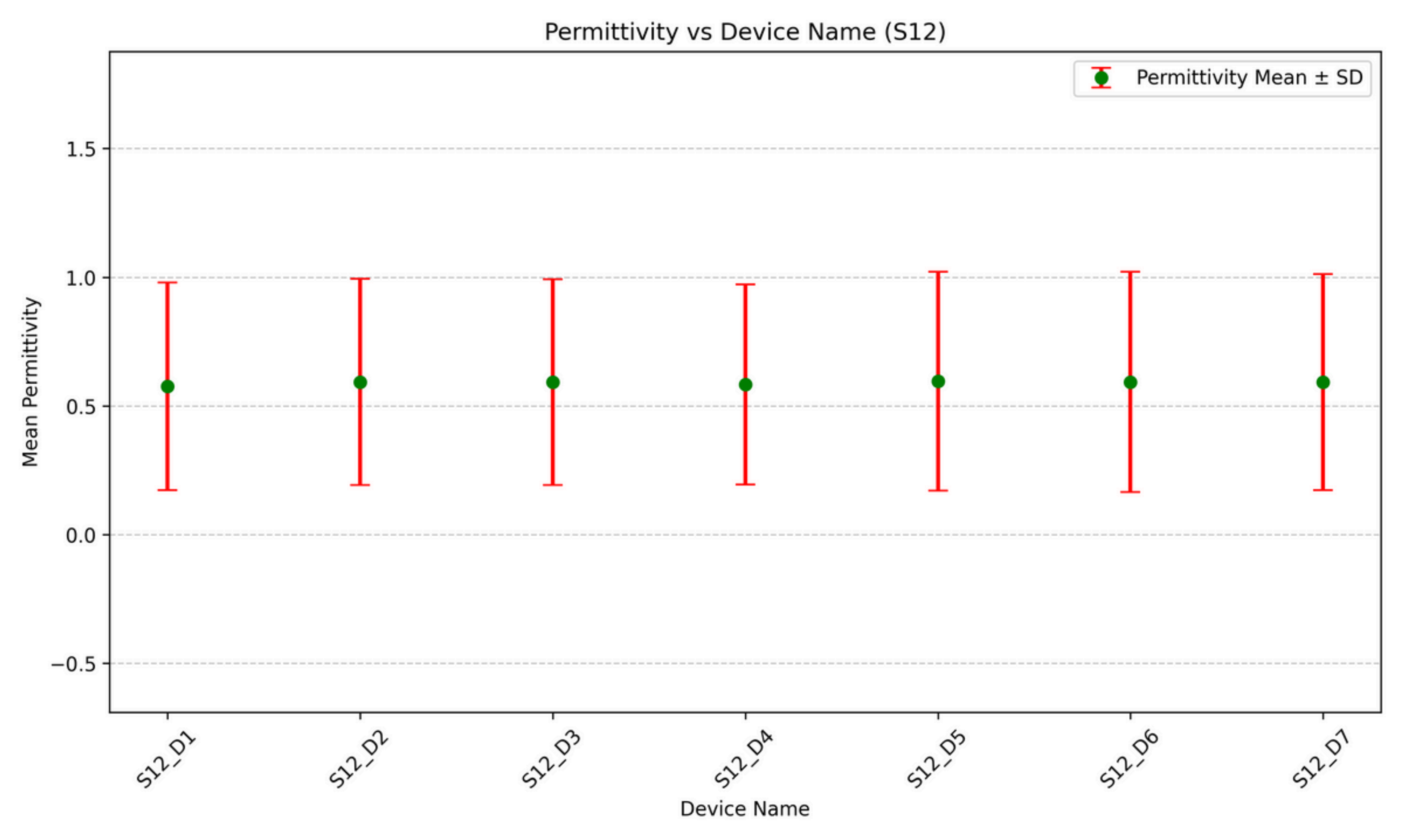

Capacitance (Cp) vs Device Name (S11)
×10⁻⁷
Cp Mean ± SD
Mean Cp (F/sq cm)
Device Name
S11_D1
S11_D2
S11_D3
S11_D4
S11_D5
S11_D6
S11_D7


Capacitance (Cp) vs Device Name (S12)
×10⁻⁹
Cp Mean ± SD
Mean Cp (F/sq cm)
Device Name
S12_D1
S12_D2
S12_D3
S12_D4
S12_D5
S12_D6
S12_D7

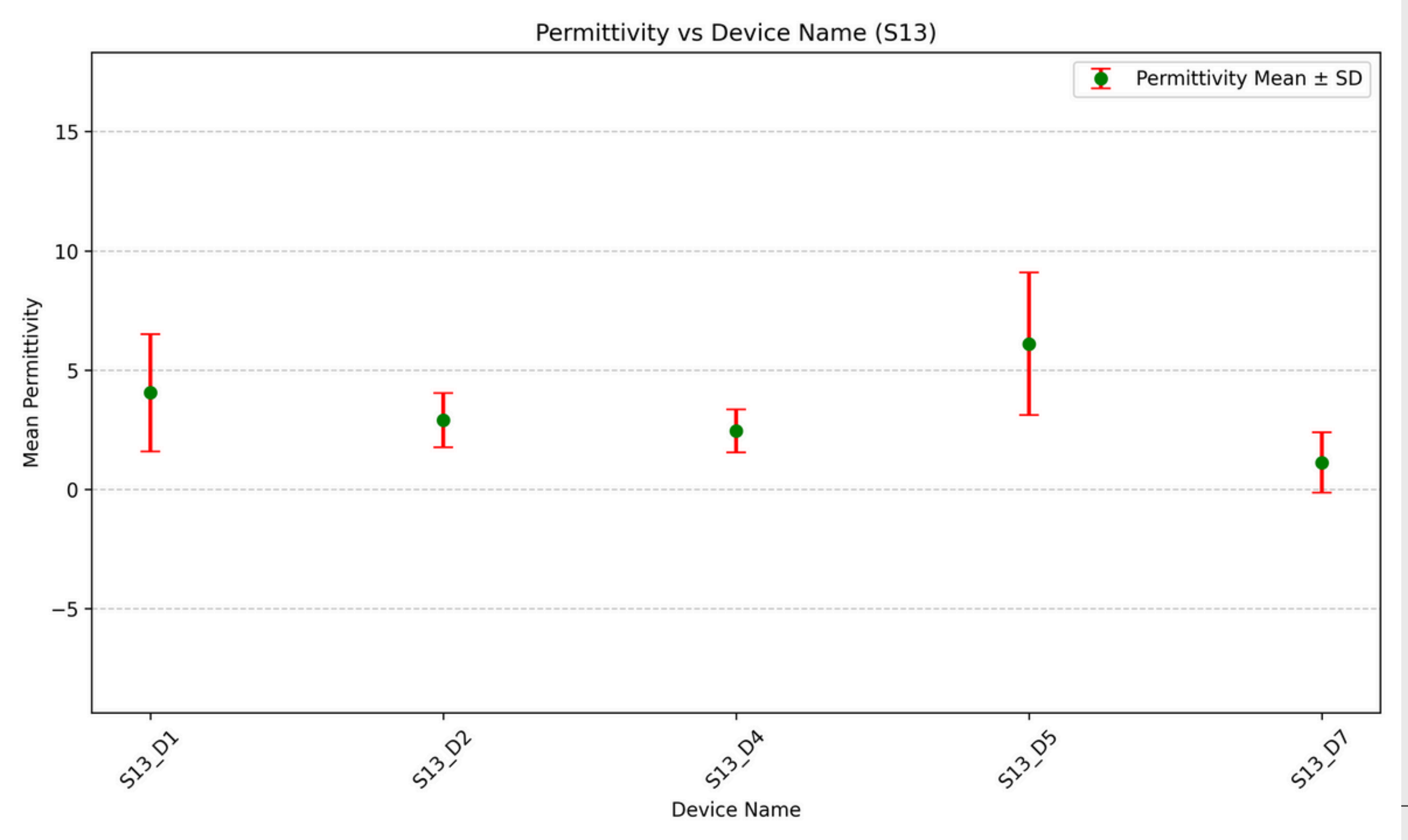

Capacitance (Cp) vs Device Name (S13)
×10⁻⁸
Cp Mean ± SD
Mean Cp (F/sq cm)
Device Name
S13_D1
S13_D2
S13_D4
S13_D5
S13_D7



Permittivity vs Device Name (S11)
Permittivity Mean ± SD
Mean Permittivity
150
100
50
0
−50
−100
S11_D1
S11_D2
S11_D3
S11_D4
S11_D5
S11_D6
S11_D7
Device Name



Permittivity vs Device Name (S12)
Permittivity Mean ± SD
Mean Permittivity
1.5
1.0
0.5
0.0
−0.5
S12_D1
S12_D2
S12_D3
S12_D4
S12_D5
S12_D6
S12_D7
Device Name



Permittivity vs Device Name (S13)
Permittivity Mean ± SD
Mean Permittivity
15
10
5
0
−5
S13_D1
S13_D2
S13_D4
S13_D5
S13_D7
Device Name

PFM = Piezoresponse Force Microscopy

PFM is a scanning probe microscopy technique (a mode of Atomic Force Microscopy, AFM) used to visualize and measure piezoelectric or ferroelectric properties at the nanoscale.

How Does PFM Work?

1. A conductive AFM tip is brought into contact with the sample surface.
2. An AC voltage is applied between the tip and the sample.
3. This causes the piezoelectric material under the tip to mechanically vibrate due to the inverse piezoelectric effect.
4. The tip detects this vertical or lateral vibration (due to d33 or d31 piezoelectric coefficients).
5. The vibration amplitude and phase are measured to map local piezoresponse.

Dataset Format

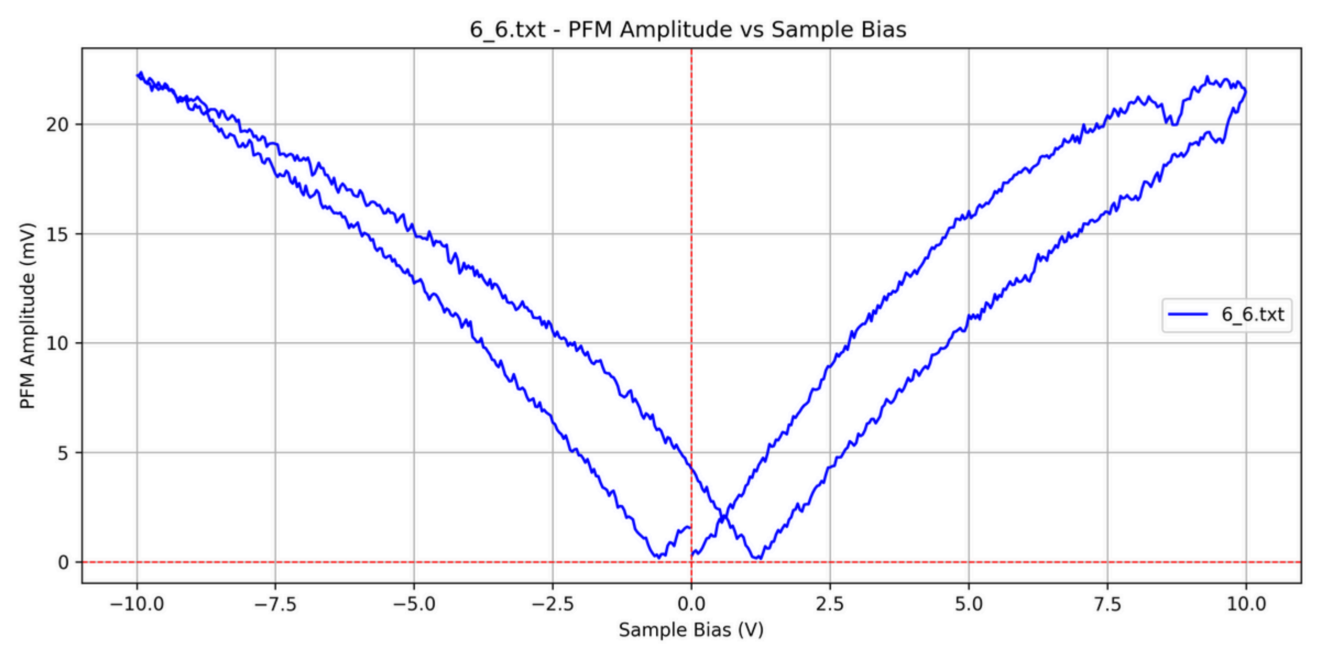
point 1

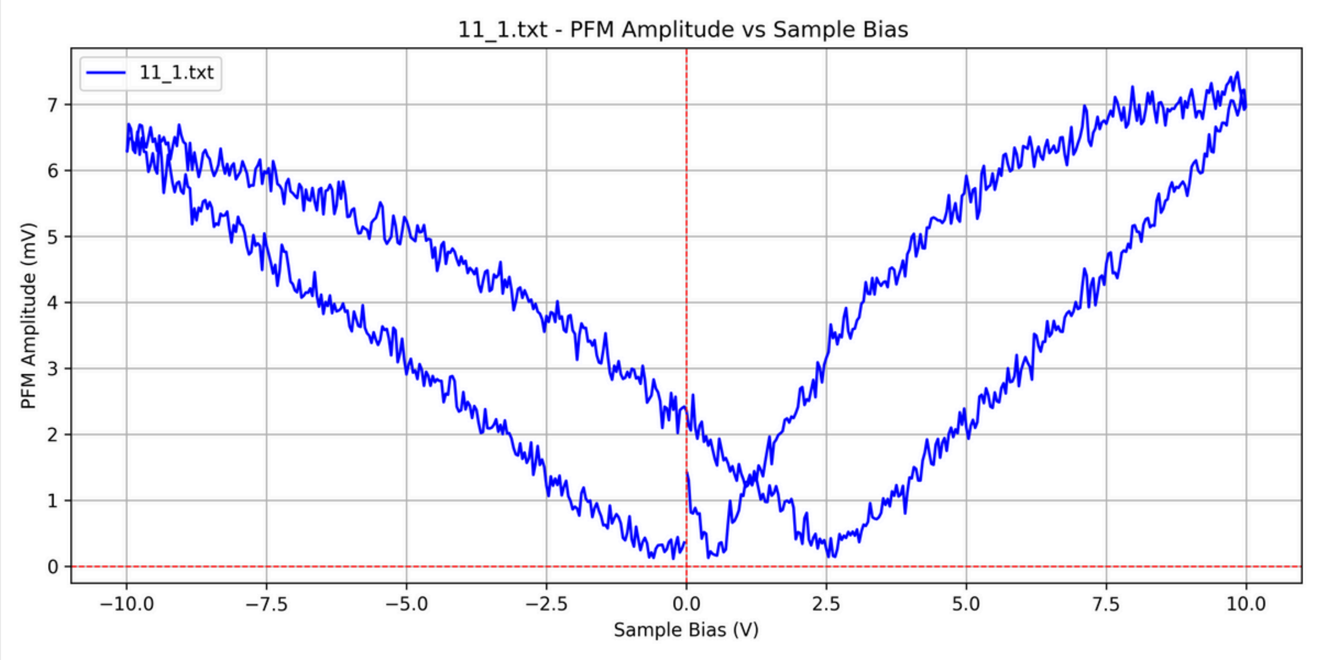
Index	Sample Bias	PFM Amplitude	PFM Phase	PFM Quad
	V V	deg V		
1	9.7591E-3	7.9956E-4	8.9289E1	3.9661E-4
2	3.9118E-2	4.3379E-4	9.9490E1	2.0712E-4
3	7.8158E-2	5.8517E-4	4.3949E1	2.0215E-4
4	1.1720E-1	3.8460E-4	3.5342E1	1.2554E-4
5	1.5624E-1	3.0431E-4	-6.5797E1	-1.4082E-4
6	1.9528E-1	7.5067E-4	-1.0537E2	-3.5365E-4
7	2.3432E-1	1.0824E-3	-1.1843E2	-4.7198E-4
8	2.7344E-1	1.2046E-3	-1.0180E2	-5.8726E-4
9	3.1256E-1	1.3241E-3	-1.0289E2	-6.4087E-4
10	3.5160E-1	1.6182E-3	-1.1050E2	-7.5851E-4
11	3.9064E-1	2.1434E-3	-1.1068E2	-9.9872E-4
12	4.2968E-1	2.4058E-3	-1.1040E2	-1.1266E-3
13	4.6872E-1	2.9300E-3	-1.0508E2	-1.4139E-3
14	5.0776E-1	2.7924E-3	-1.0624E2	-1.3403E-3
15	5.4688E-1	3.5769E-3	-1.0233E2	-1.7469E-3
16	5.8600E-1	3.5161E-3	-1.0911E2	-1.6606E-3
17	6.2504E-1	3.7610E-3	-1.0766E2	-1.7918E-3
18	6.6408E-1	3.7997E-3	-1.0919E2	-1.7946E-3
19	7.0312E-1	4.1829E-3	-1.1126E2	-1.9485E-3
20	7.4216E-1	4.5198E-3	-1.1084E2	-2.1126E-3

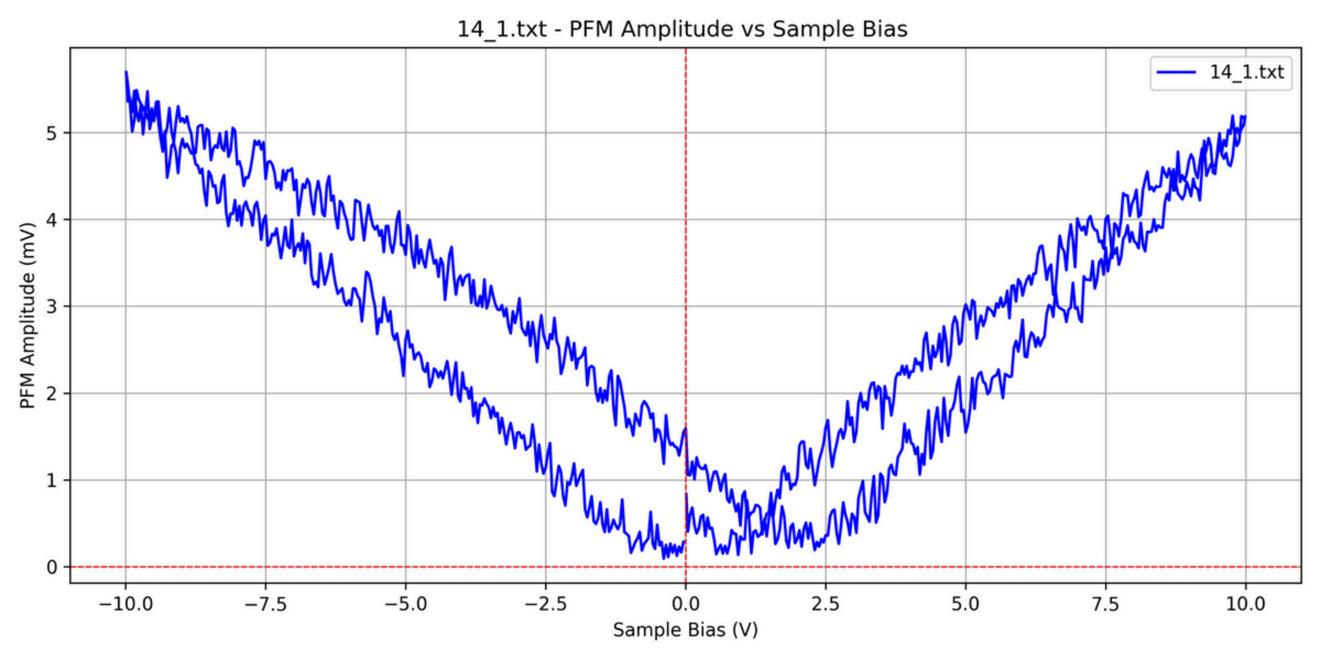
Code for pfm plots

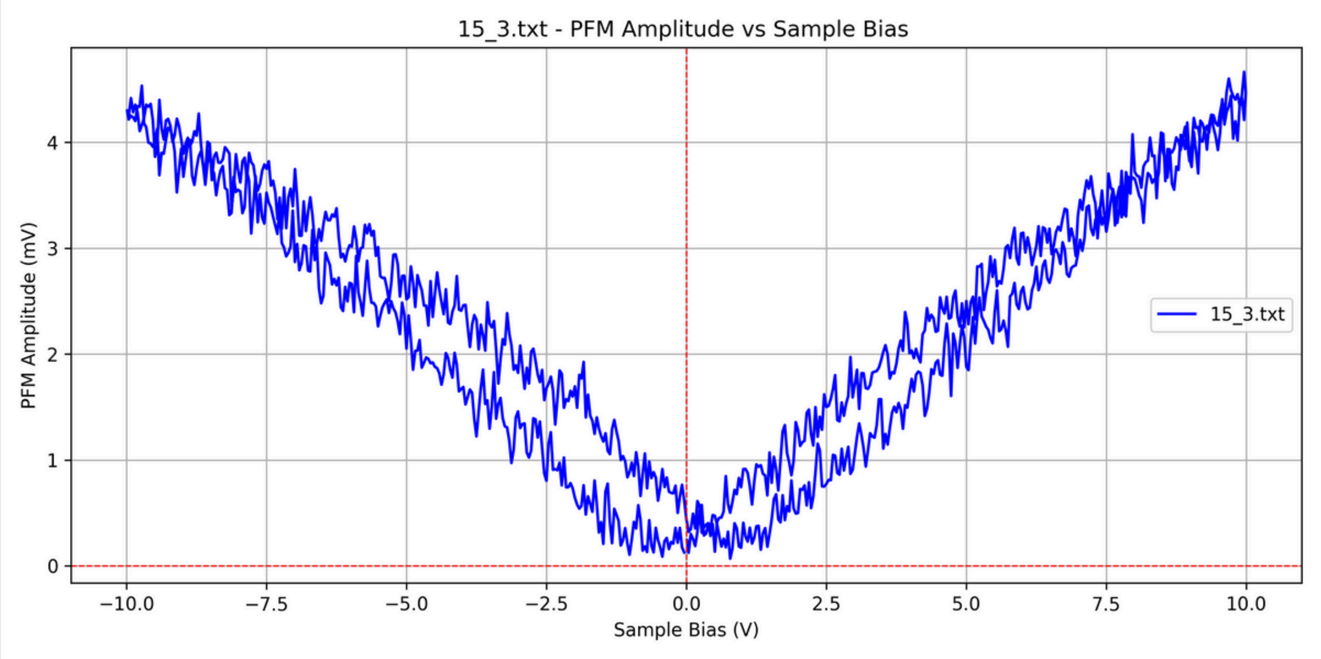
```python
folder_path = '/Users/asif/Desktop/PFM data'

# Loop through all .txt files in the folder
for filename in os.listdir(folder_path):
    if filename.endswith('.txt'):
        file_path = os.path.join(folder_path, filename)
        print(f"Processing: {filename}")

        try:
            # Read and clean the data
            data = pd.read_csv(file_path, sep='\t', skiprows=1)
            data.columns = data.columns.str.strip()

            # Convert to numeric
            y1=data['PFM Amplitude'] = pd.to_numeric(data['PFM Amplitude'], errors='coerce')
            y=y1*1000
            x=data['Sample Bias'] = pd.to_numeric(data['Sample Bias'], errors='coerce')
            data.dropna(subset=['PFM Amplitude', 'Sample Bias'], inplace=True)

            # Plot
            plt.figure(figsize=(10, 5))
            plt.plot( x,y, label=filename, color='blue')
            plt.ylabel('PFM Amplitude (mV)')
            plt.xlabel('Sample Bias (V)')
            plt.yscale('linear')
            plt.xscale('linear')
            plt.title(f'{filename} - PFM Amplitude vs Sample Bias ')
            plt.grid(True)
            plt.tight_layout()
            plt.legend()
            plt.axhline(0, color='red', linewidth=0.8, linestyle='--')  # X-axis
            plt.axvline(0, color='red', linewidth=0.8, linestyle='--')  # Y-axis
            # Save plot
            plot_filename = os.path.splitext(filename)[0] + '_plot.png'
            plt.savefig(os.path.join(folder_path, plot_filename), dpi=300)
            # plt.show()
            plt.close()
```

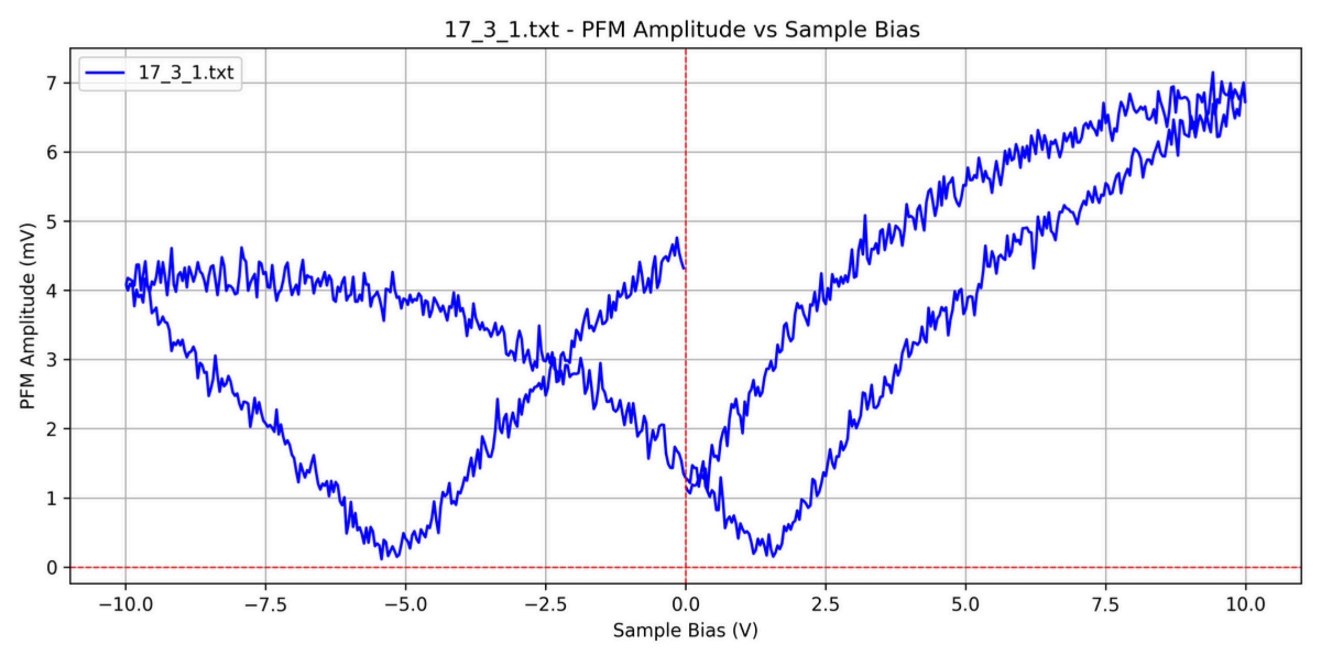
PFM plots: https://drive.google.com/file/d/12UladNx9j1vG1X5uBAhpd6VdNLdc-dnH/view?usp=sharing

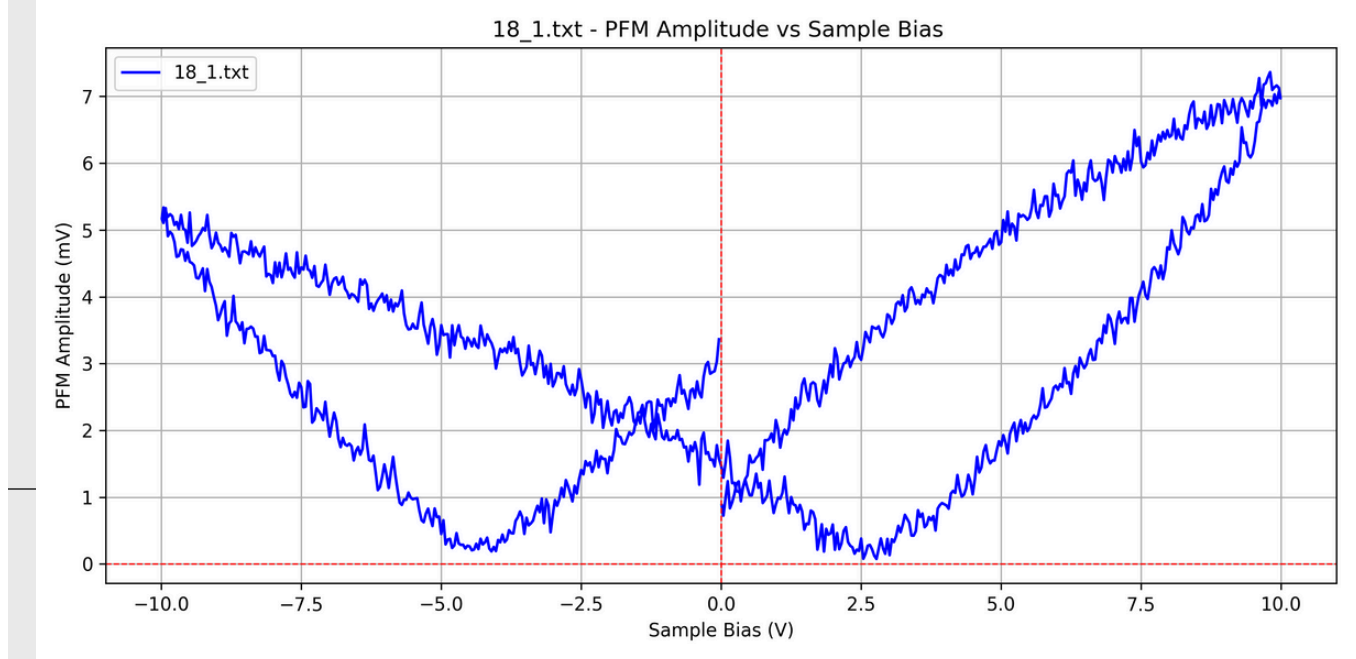

6_6.txt - PFM Amplitude vs Sample Bias
PFM Amplitude (mV)
Sample Bias (V)
6_6.txt


11_1.txt - PFM Amplitude vs Sample Bias
PFM Amplitude (mV)
Sample Bias (V)
11_1.txt


14_1.txt - PFM Amplitude vs Sample Bias
PFM Amplitude (mV)
Sample Bias (V)
14_1.txt


15_3.txt - PFM Amplitude vs Sample Bias
PFM Amplitude (mV)
Sample Bias (V)
15_3.txt


17_3_1.txt - PFM Amplitude vs Sample Bias
PFM Amplitude (mV)
Sample Bias (V)
17_3_1.txt


18_1.txt - PFM Amplitude vs Sample Bias
PFM Amplitude (mV)
Sample Bias (V)
18_1.txt

What is d_{33}?

d_{33} is the piezoelectric charge coefficient (or strain coefficient) that describes the longitudinal piezoelectric response — i.e., how much the material expands or contracts along the same direction as the applied electric field.

$$d_{33} = \frac{\text{displacement (m)}}{\text{electric field (V/m)}} = \text{m/V} \quad \text{(usually reported in pm/V)}$$

How to calculate d33?

Slope of displacement vs sample bias(volts) curve gives d33.

Dataset Format

point 1

Index	Sample Bias	PFM Amplitude	PFM Phase	PFM Quad
	V V	deg V		
1	9.7591E-3	7.9956E-4	8.9289E1	3.9661E-4
2	3.9118E-2	4.3379E-4	9.9490E1	2.0712E-4
3	7.8158E-2	5.8517E-4	4.3949E1	2.0215E-4
4	1.1720E-1	3.8460E-4	3.5342E1	1.2554E-4
5	1.5624E-1	3.0431E-4	-6.5797E1	-1.4082E-4
6	1.9528E-1	7.5067E-4	-1.0537E2	-3.5365E-4
7	2.3432E-1	1.0824E-3	-1.1843E2	-4.7198E-4
8	2.7344E-1	1.2046E-3	-1.0180E2	-5.8726E-4
9	3.1256E-1	1.3241E-3	-1.0289E2	-6.4087E-4
10	3.5160E-1	1.6182E-3	-1.1050E2	-7.5851E-4
11	3.9064E-1	2.1434E-3	-1.1068E2	-9.9872E-4
12	4.2968E-1	2.4058E-3	-1.1040E2	-1.1266E-3
13	4.6872E-1	2.9300E-3	-1.0508E2	-1.4139E-3
14	5.0776E-1	2.7924E-3	-1.0624E2	-1.3403E-3
15	5.4688E-1	3.5769E-3	-1.0233E2	-1.7469E-3
16	5.8600E-1	3.5161E-3	-1.0911E2	-1.6606E-3
17	6.2504E-1	3.7610E-3	-1.0766E2	-1.7918E-3
18	6.6408E-1	3.7997E-3	-1.0919E2	-1.7946E-3
19	7.0312E-1	4.1829E-3	-1.1126E2	-1.9485E-3
20	7.4216E-1	4.5198E-3	-1.1084E2	-2.1126E-3

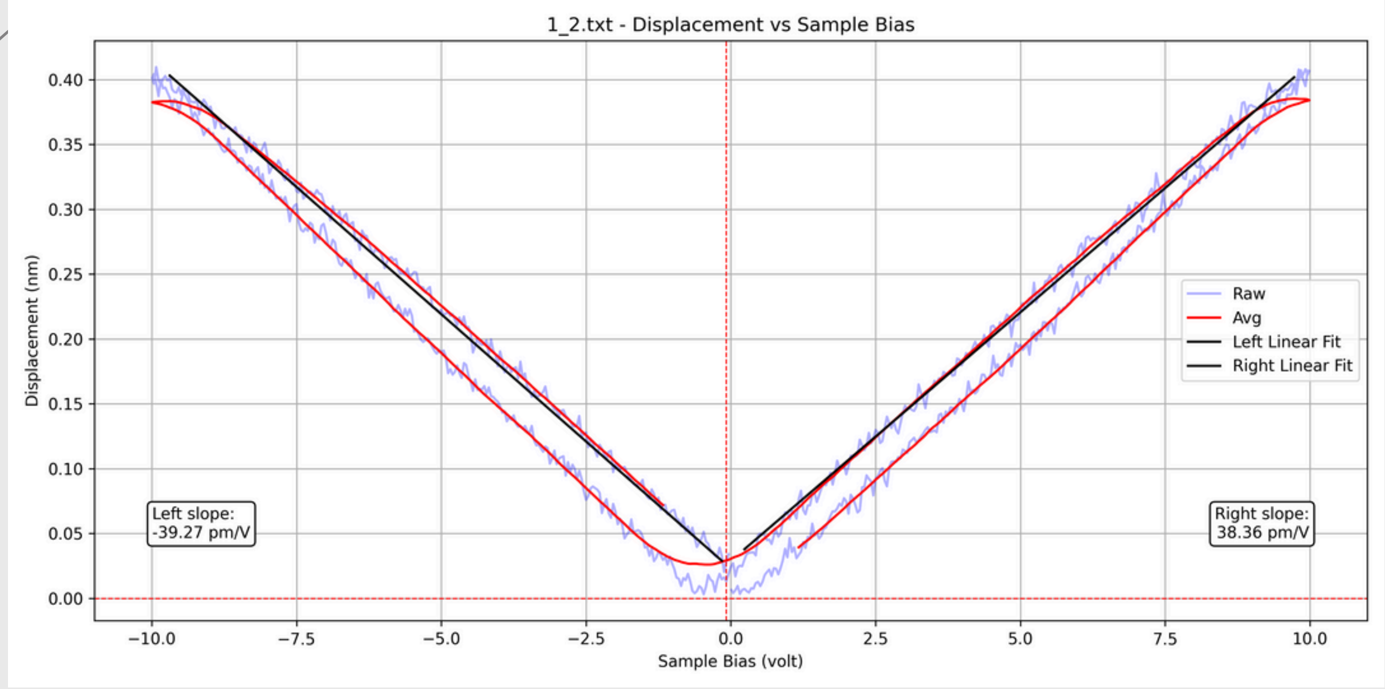
PFM Amplitude/sensitivity=displacement

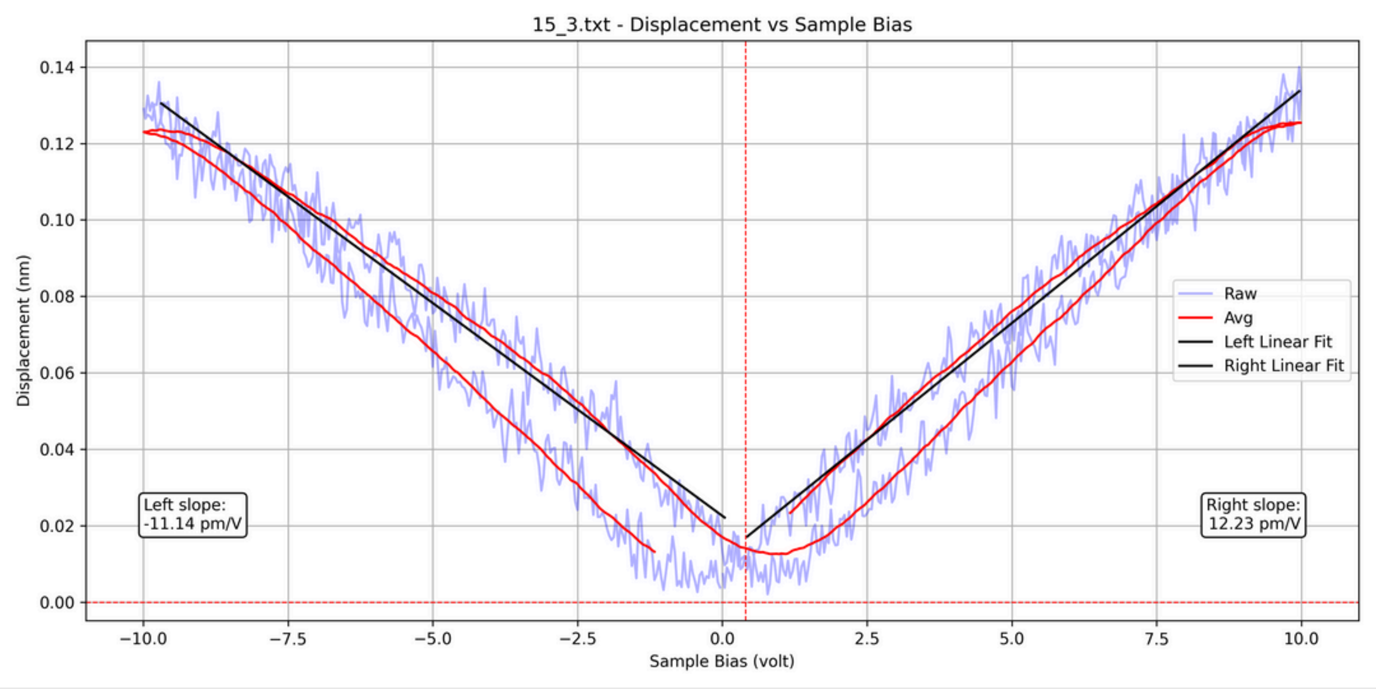
$$\frac{10^{-3}\,\mathrm{mV}}{33.333332\,\mathrm{V}\cdot\mu\mathrm{m}^{-1}} = \frac{10^{-9}\,\mathrm{m}}{33.333332} = \frac{1\,\mathrm{nm}}{33.333332}$$

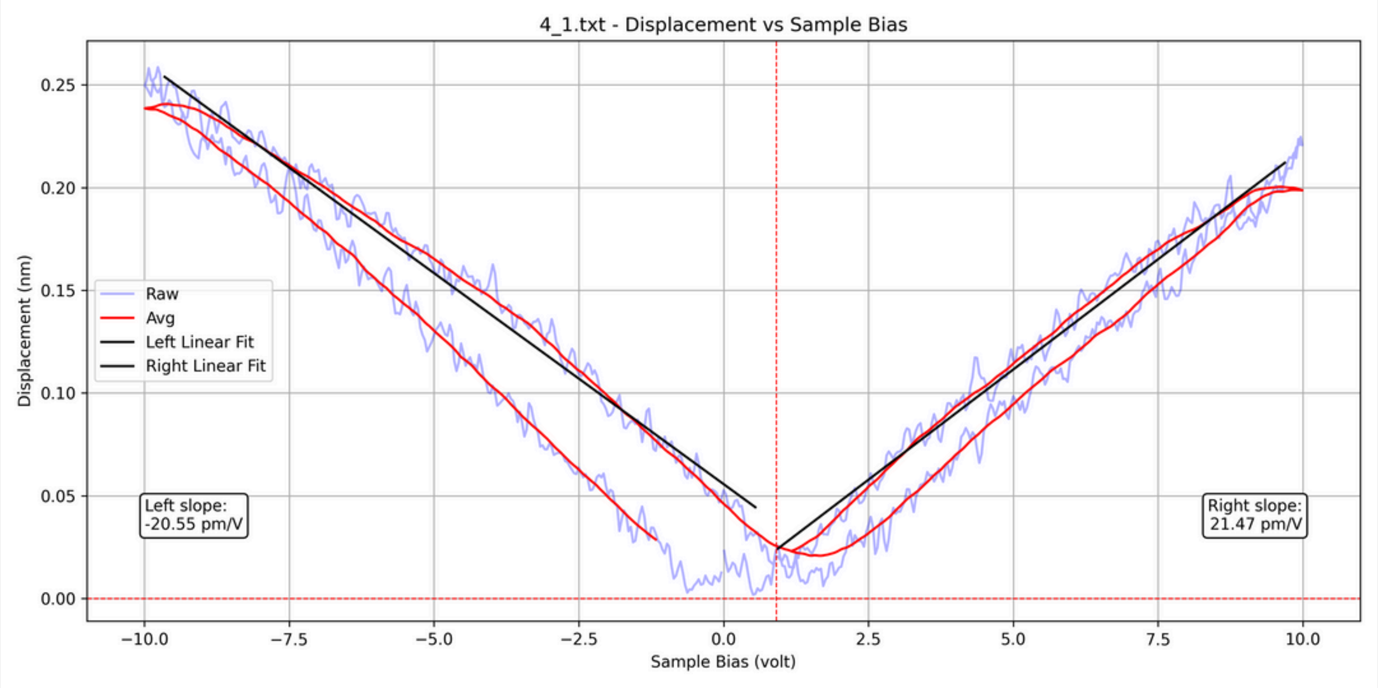
Code for displacement vs sample bias

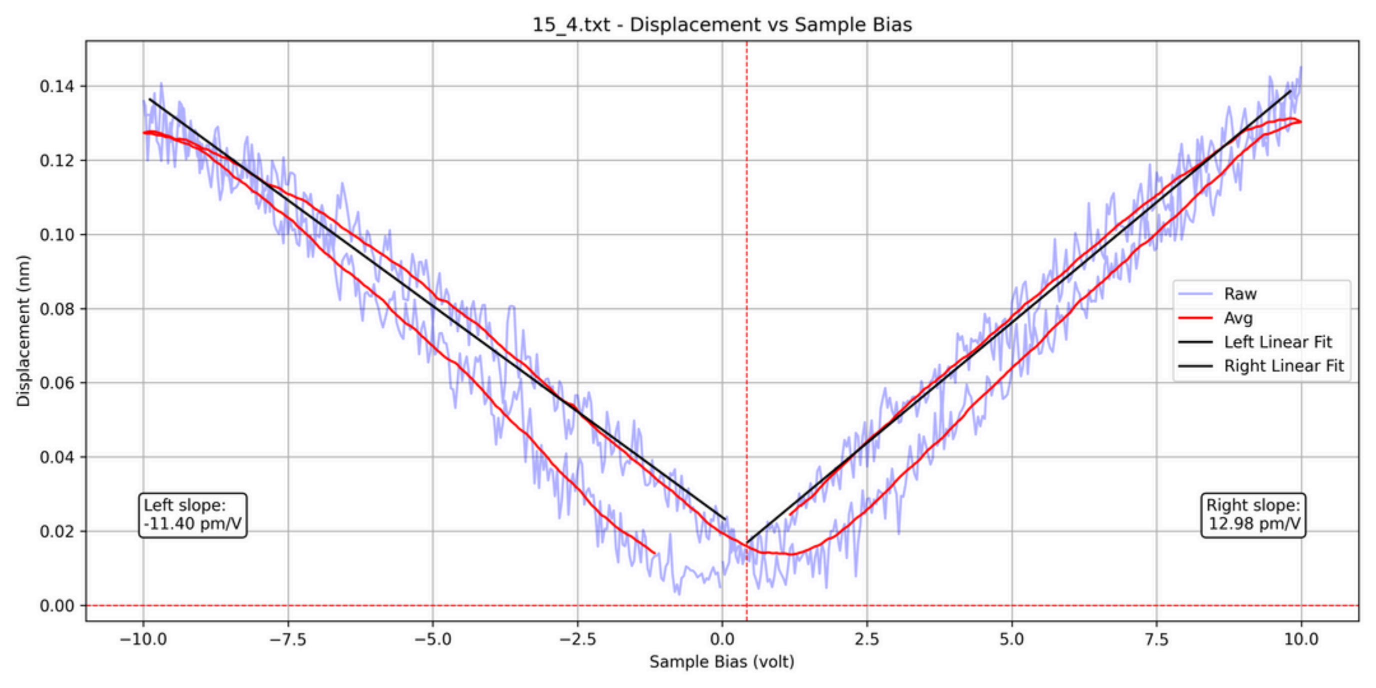
```python
data = pd.read_csv(file_path, sep='\t', skiprows=1)
data.columns = data.columns.str.strip()
data['PFM Amplitude'] = pd.to_numeric(data['PFM Amplitude'], errors='coerce')
data['Sample Bias'] = pd.to_numeric(data['Sample Bias'], errors='coerce')
data.dropna(subset=['PFM Amplitude', 'Sample Bias'], inplace=True)

x = data['Sample Bias']
y = data['PFM Amplitude']*1000/33.333332
y_avg = y.rolling(window=60, center=True).mean()

split_x = find_intersection(x, y_avg)
print(f"Intersection at x = {split_x:.3f} V")

plt.figure(figsize=(12, 6))
plt.plot(x, y, label='Raw', color='blue', alpha=0.3)
plt.plot(x, y_avg, label='Avg', color='red')

left_slope = fit_and_plot_max_line(x, y_avg, region='left', split_x=split_x, color='black', label='Left Linear Fit')
right_slope = fit_and_plot_max_line(x, y_avg, region='right', split_x=split_x, color='black', label='Right Linear Fit')

slope_data['filenames'].append(filename)
slope_data['left_slope'].append(left_slope * 1000)
slope_data['right_slope'].append(right_slope * 1000)
slope_data['intersection'].append(split_x)

plt.axhline(0, color='red', linestyle='--', linewidth=0.8)
plt.axvline(split_x, color='red', linestyle='--', linewidth=0.8)
plt.xlabel('Sample Bias (volt)')
plt.ylabel('Displacement (nm)')
plt.title(f'{filename} - Displacement vs Sample Bias')
plt.legend()
plt.grid(True)
plt.tight_layout()

save_path = os.path.join(output_plot_path, f'{filename[:-4]}.png')
plt.savefig(save_path, dpi=300)
plt.close()
print(f"Plot saved to: {save_path}")
```

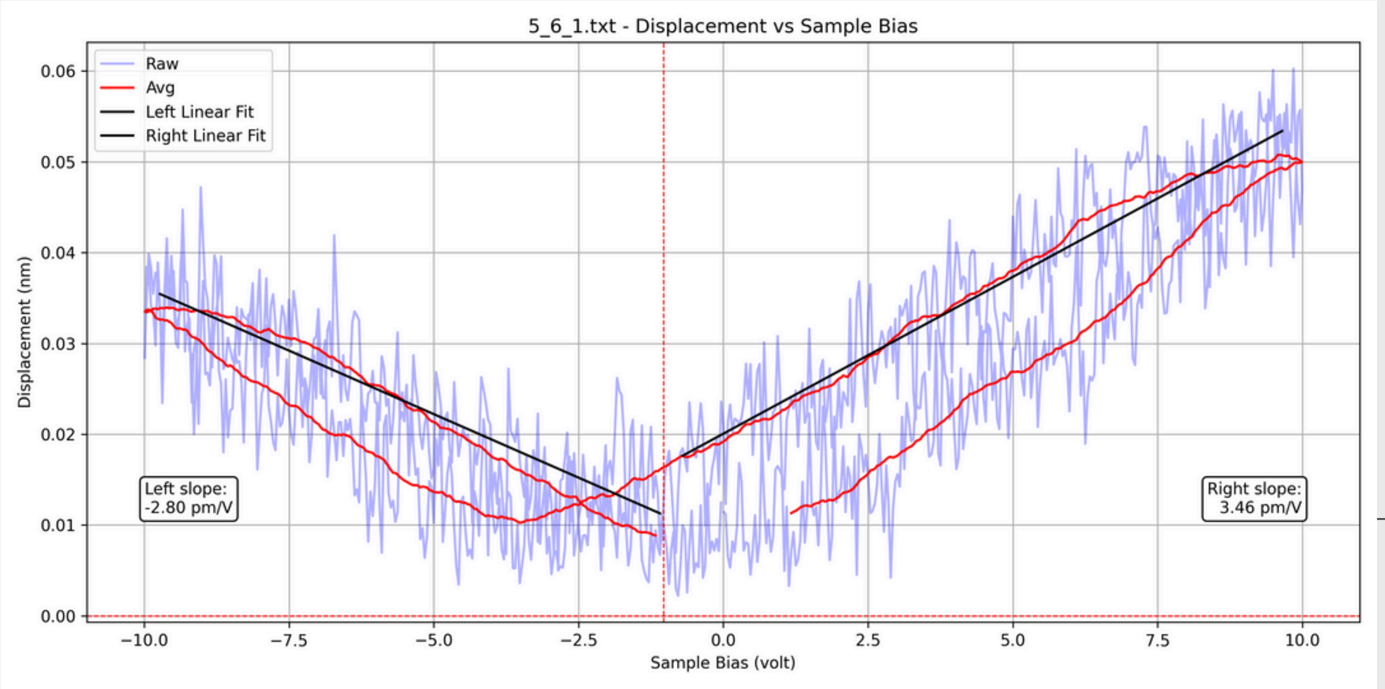
d33 calculation: https://drive.google.com/file/d/12Cyn9Ga3Ccr-Ry6bjPELuZiwxz8ZkNZC/view?usp=sharing

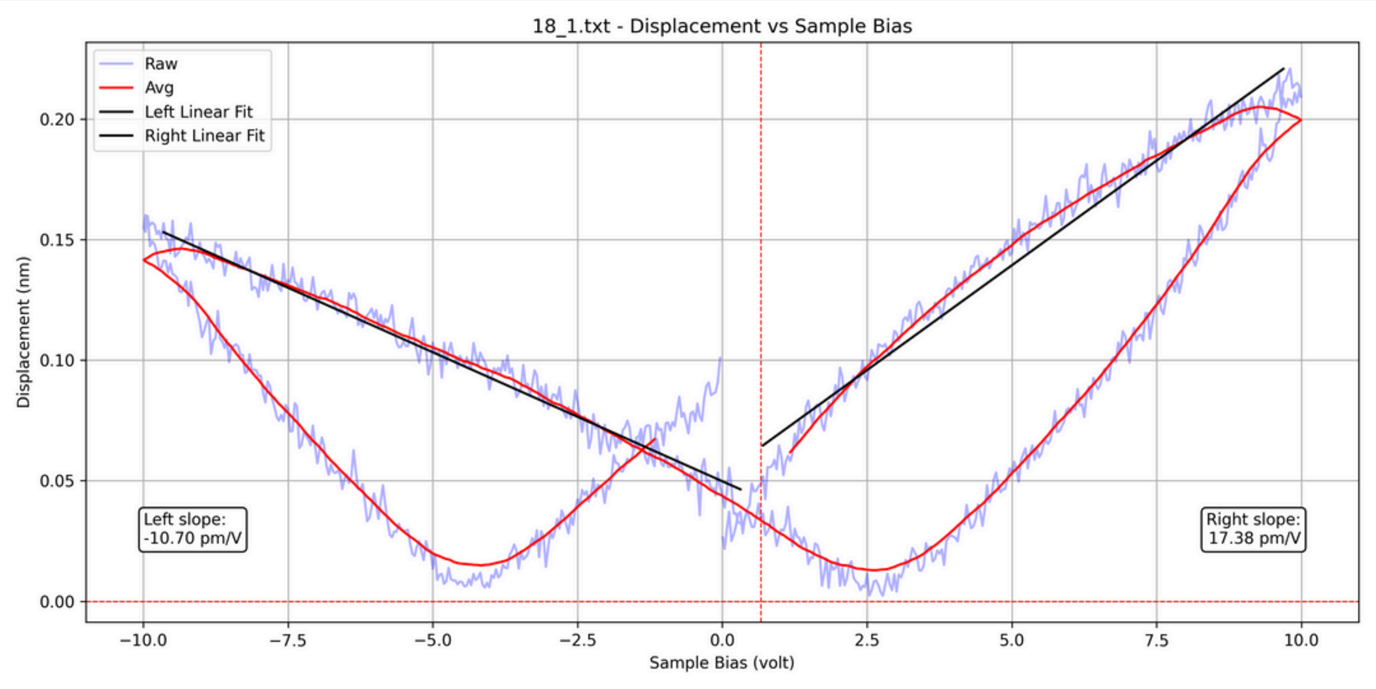

1_2.txt - Displacement vs Sample Bias
Raw
Avg
Left Linear Fit
Right Linear Fit
Left slope: -39.27 pm/V
Right slope: 38.36 pm/V
Displacement (nm)
Sample Bias (volt)


15_3.txt - Displacement vs Sample Bias
Raw
Avg
Left Linear Fit
Right Linear Fit
Left slope: -11.14 pm/V
Right slope: 12.23 pm/V
Displacement (nm)
Sample Bias (volt)


4_1.txt - Displacement vs Sample Bias
Raw
Avg
Left Linear Fit
Right Linear Fit
Left slope: -20.55 pm/V
Right slope: 21.47 pm/V
Displacement (nm)
Sample Bias (volt)


15_4.txt - Displacement vs Sample Bias
Raw
Avg
Left Linear Fit
Right Linear Fit
Left slope: -11.40 pm/V
Right slope: 12.98 pm/V
Displacement (nm)
Sample Bias (volt)


5_6_1.txt - Displacement vs Sample Bias
Raw
Avg
Left Linear Fit
Right Linear Fit
Left slope: -2.80 pm/V
Right slope: 3.46 pm/V
Displacement (nm)
Sample Bias (volt)


18_1.txt - Displacement vs Sample Bias
Raw
Avg
Left Linear Fit
Right Linear Fit
Left slope: -10.70 pm/V
Right slope: 17.38 pm/V
Displacement (nm)
Sample Bias (volt)

Position 1				
filenames	position name	left_slope	right_slope	intersection
1_1.txt	1_1	-47.492 pm/V	254.5524 pm/V	0.9439835564
1_2.txt	1_2	-39.2717 pm/V	38.3582 pm/V	-0.08554409639
Mean Left Slope	-43.3819 pm/V			
Mean Right Slope	146.4553 pm/V			

Position 2				
filenames	position name	left_slope	right_slope	intersection
2_1.txt	2_1	-15.0632 pm/V	16.6954 pm/V	-0.00108323285
2_2.txt	2_2	-4.7697 pm/V	4.3079 pm/V	1.40592331
2_3.txt	2_3	-2.2222 pm/V	2.4631 pm/V	-1.25252143
2_4.txt	2_4	-118.6614 pm/V	116.8124 pm/V	-0.1944837695
Mean Left Slope	-35.1791 pm/V			
Mean Right Slope	35.0697 pm/V			

Position 3				
filenames	position name	left_slope	right_slope	intersection
3_1.txt	3_1	-20.9148 pm/V	17.0129 pm/V	0.7901055122
3_2.txt	3_2	-163.8954 pm/V	162.7572 pm/V	-0.1154694849
Mean Left Slope	-92.4051 pm/V			
Mean Right Slope	89.8851 pm/V			

Position 4				
filenames	position name	left_slope	right_slope	intersection
4_1.txt	4_1	-20.5505 pm/V	21.4675 pm/V	0.9073875548
4_2.txt	4_2	-8.9371 pm/V	9.017 pm/V	-0.1418921384
Mean Left Slope	-14.7438 pm/V			
Mean Right Slope	15.2423 pm/V			

Position 5				
filenames	position name	left_slope	right_slope	intersection
5_1_1.txt	5_1	-17.4207 pm/V	16.9851 pm/V	-0.09621031839
5_1.txt	5_1	-18.2849 pm/V	19.0803 pm/V	-0.1141605989
5_2_1.txt	5_2	-21.19 pm/V	22.1995 pm/V	0.4889283753
5_3_1.txt	5_3	-13.6152 pm/V	15.0358 pm/V	0.8831491406
5_3_2.txt	5_3	-135.7515 pm/V	130.6562 pm/V	-0.099345039
5_4_1.txt	5_4	-0.7034 pm/V	0.7485 pm/V	0.1853579108
5_5_1.txt	5_5	-0.7896 pm/V	0.724 pm/V	-0.08459318331
5_6_1.txt	5_6	-2.7987 pm/V	3.4552 pm/V	-1.032461269
5_7_1.txt	5_7	-5.1597 pm/V	6.4528 pm/V	-1.16939481
Mean Left Slope	-23.9682 pm/V			
Mean Right Slope	23.9264 pm/V			

Position 6				
filenames	position name	left_slope	right_slope	intersection
6_1_1.txt	6_1	-38.1965 pm/V	114.1777 pm/V	0.7885050771
6_1.txt	6_1	-99.3955 pm/V	87.3065 pm/V	0.3514036256
6_2.txt	6_2	-2.6658 pm/V	2.5449 pm/V	0.6233261006
6_3.txt	6_3	-40.4091 pm/V	44.0589 pm/V	-0.1781752635
6_4.txt	6_4	-17.2322 pm/V	17.4212 pm/V	-0.02203468428
6_5.txt	6_5	-22.5231 pm/V	23.2685 pm/V	-0.342588452
6_6.txt	6_6	-52.7608 pm/V	62.2146 pm/V	0.5774686833
Mean Left Slope	-39.0261 pm/V			
Mean Right Slope	50.1418 pm/V			

Position 8				
filenames	position name	left_slope	right_slope	intersection
8_1.txt	8_1	-98.5856 pm/V	106.2551 pm/V	-0.0450513931
8_1_1.txt	8_1	-5.0442 pm/V	5.584 pm/V	0.5534360751
Mean Left Slope	-51.8149 pm/V			
Mean Right Slope	55.9196 pm/V			

Position 9				
filenames	position name	left_slope	right_slope	intersection
9_1.txt	9_1	-3.3014 pm/V	4.4988 pm/V	-0.05991146034
Mean Left Slope	-3.3014 pm/V			
Mean Right Slope	4.4988 pm/V			

Position 10				
filenames	position name	left_slope	right_slope	intersection
10_1.txt	10_1	-7.5068 pm/V	12.6069 pm/V	-0.3392957714
10_2.txt	10_2	-7.1932 pm/V	6.8359 pm/V	0.495775802
10_3.txt	10_3	-23.5193 pm/V	29.646 pm/V	0.5035979922
10_4.txt	10_4	-6.0921 pm/V	10.4876 pm/V	0.710916958
Mean Left Slope	-11.0779 pm/V			
Mean Right Slope	14.8941 pm/V			

Position 11				
filenames	position name	left_slope	right_slope	intersection
11_1.txt	11_1	-13.1926 pm/V	19.2352 pm/V	0.9255997058
Mean Left Slope	-13.1926 pm/V			
Mean Right Slope	19.2352 pm/V			

Position 12				
filenames	position name	left_slope	right_slope	intersection
12_1.txt	12_1	-10.3918 pm/V	11.006 pm/V	0.9198906941
12_1_1.txt	12_1	-10.6371 pm/V	11.1562 pm/V	0.6771207437

12_1_1.txt	12_1	-10.6371 pm/V	11.1562 pm/V	0.6771207437

Mean Left Slope	-10.5145 pm/V			
Mean Right Slope	11.0811 pm/V			

Position 13				
filenames	position name	left_slope	right_slope	intersection
13_1.txt	13_1	-8.9359 pm/V	11.2217 pm/V	1.002613198
Mean Left Slope	-8.9359 pm/V			
Mean Right Slope	11.2217 pm/V			

Position 14				
filenames	position name	left_slope	right_slope	intersection
14_1.txt	14_1	-12.3384 pm/V	14.7179 pm/V	1.022765205
Mean Left Slope	-12.3384 pm/V			
Mean Right Slope	14.7179 pm/V			

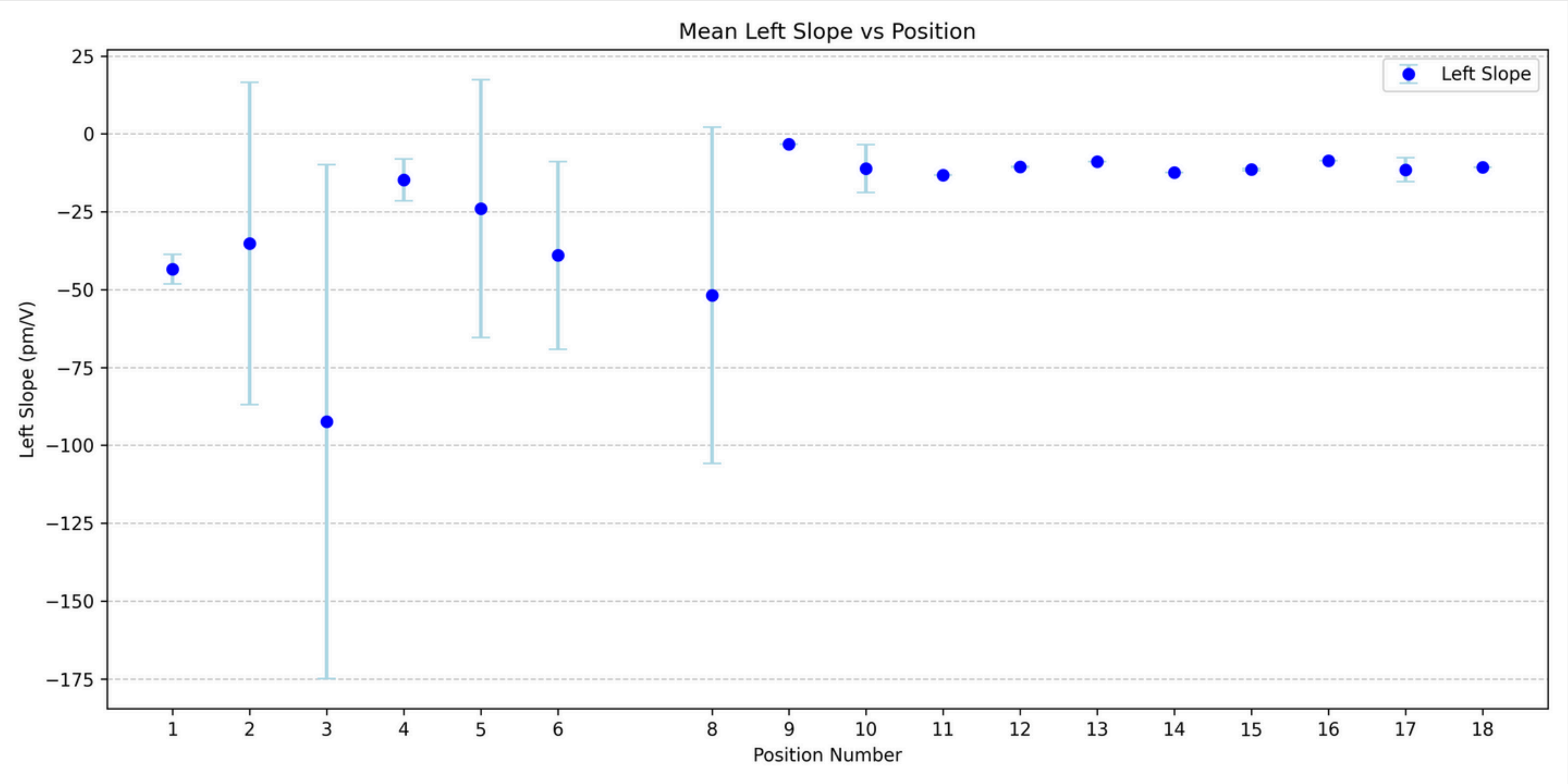
Position 15				
filenames	position name	left_slope	right_slope	intersection
15_1.txt	15_1	-11.3421 pm/V	12.842 pm/V	0.7061154943
15_2.txt	15_2	-11.425 pm/V	12.0267 pm/V	0.2781038037
15_3.txt	15_3	-11.1376 pm/V	12.2256 pm/V	0.3975614795
15_4.txt	15_4	-11.4044 pm/V	12.9792 pm/V	0.4216979439
15_5.txt	15_5	-12.0017 pm/V	13.4082 pm/V	0.3677635688
Mean Left Slope	-11.4622 pm/V			
Mean Right Slope	12.6963 pm/V			

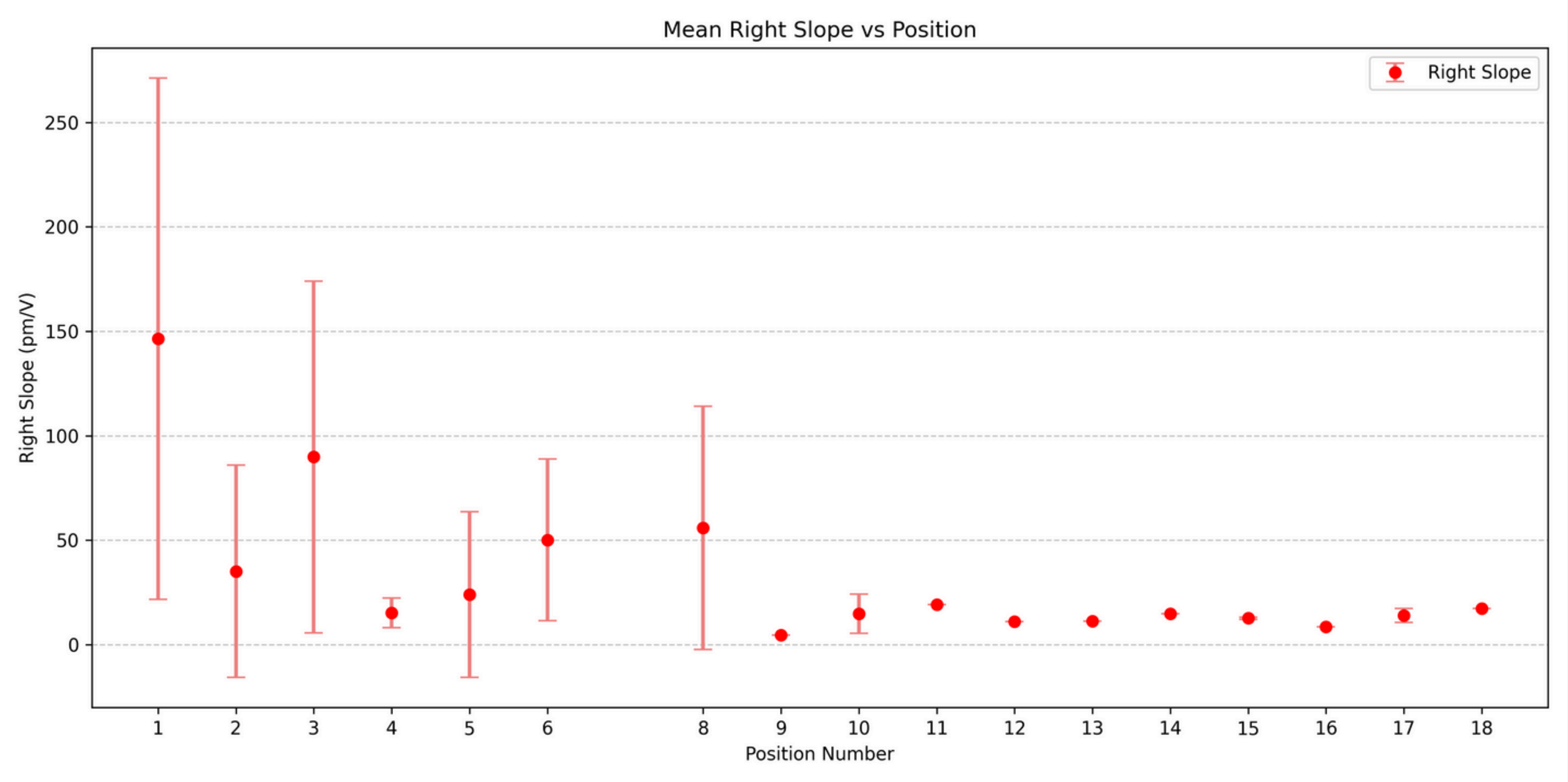
Position 16				
filenames	position name	left_slope	right_slope	intersection
16_1.txt	16_1	-8.6043 pm/V	8.6096 pm/V	0.174050522
Mean Left Slope	-8.6043 pm/V			
Mean Right Slope	8.6096 pm/V			

Position 17				
filenames	position name	left_slope	right_slope	intersection
17_1.txt	17_1	-6.8068 pm/V	9.5413 pm/V	-1.889491423
17_1_1.txt	17_1	-12.7111 pm/V	12.69 pm/V	0.4428868126
17_1_1_1.txt	17_1	-16.9011 pm/V	19.2495 pm/V	1.665601601
17_2_1.txt	17_2	-12.8975 pm/V	14.9621 pm/V	2.0644415
17_3_1.txt	17_3	-8.1507 pm/V	13.3812 pm/V	1.159839632
Mean Left Slope	-11.4934 pm/V			
Mean Right Slope	13.9648 pm/V			

Position 18				
filenames	position name	left_slope	right_slope	intersection
18_1.txt	18_1	-10.7019 pm/V	17.376 pm/V	0.6656021458
Mean Left Slope	-10.7019 pm/V			
Mean Right Slope	17.3760 pm/V			

```
# Plot Left Slope
plt.figure(figsize=(12, 6))
plt.errorbar(
    x=grouped['position'],
    y=grouped['left_mean'],
    yerr=grouped['left_std'],
    fmt='o',
    color='blue',
    ecolor='lightblue',
    elinewidth=2,
    capsize=5,
    label='Left Slope'
)
plt.title('Mean Left Slope vs Position')
plt.xlabel('Position Number')
plt.ylabel('Left Slope (pm/V)')
plt.xticks(grouped['position'])
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.legend()
plt.tight_layout()
plt.savefig('Left_Slope_vs_Position.png', dpi=300)
# plt.show()
```

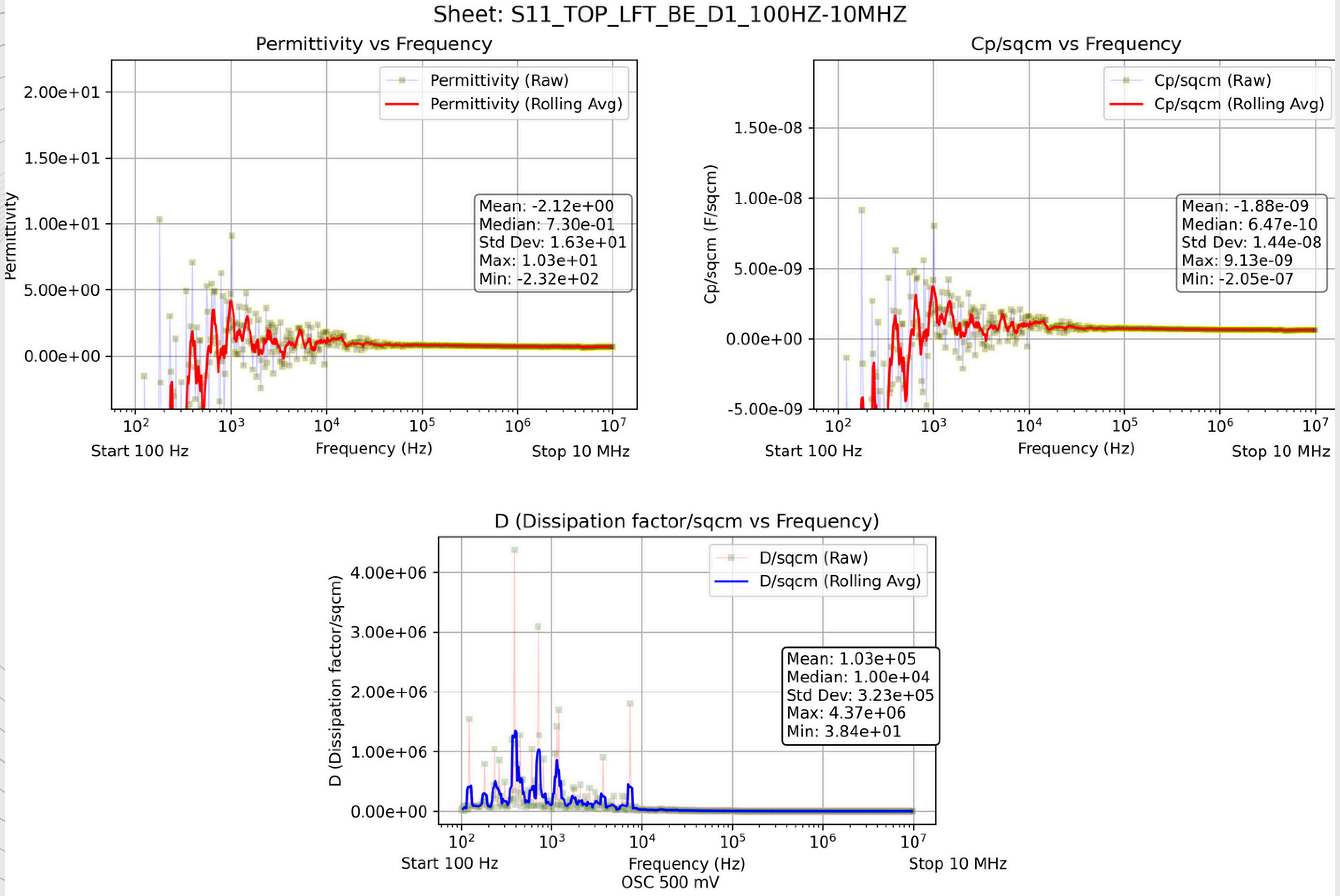


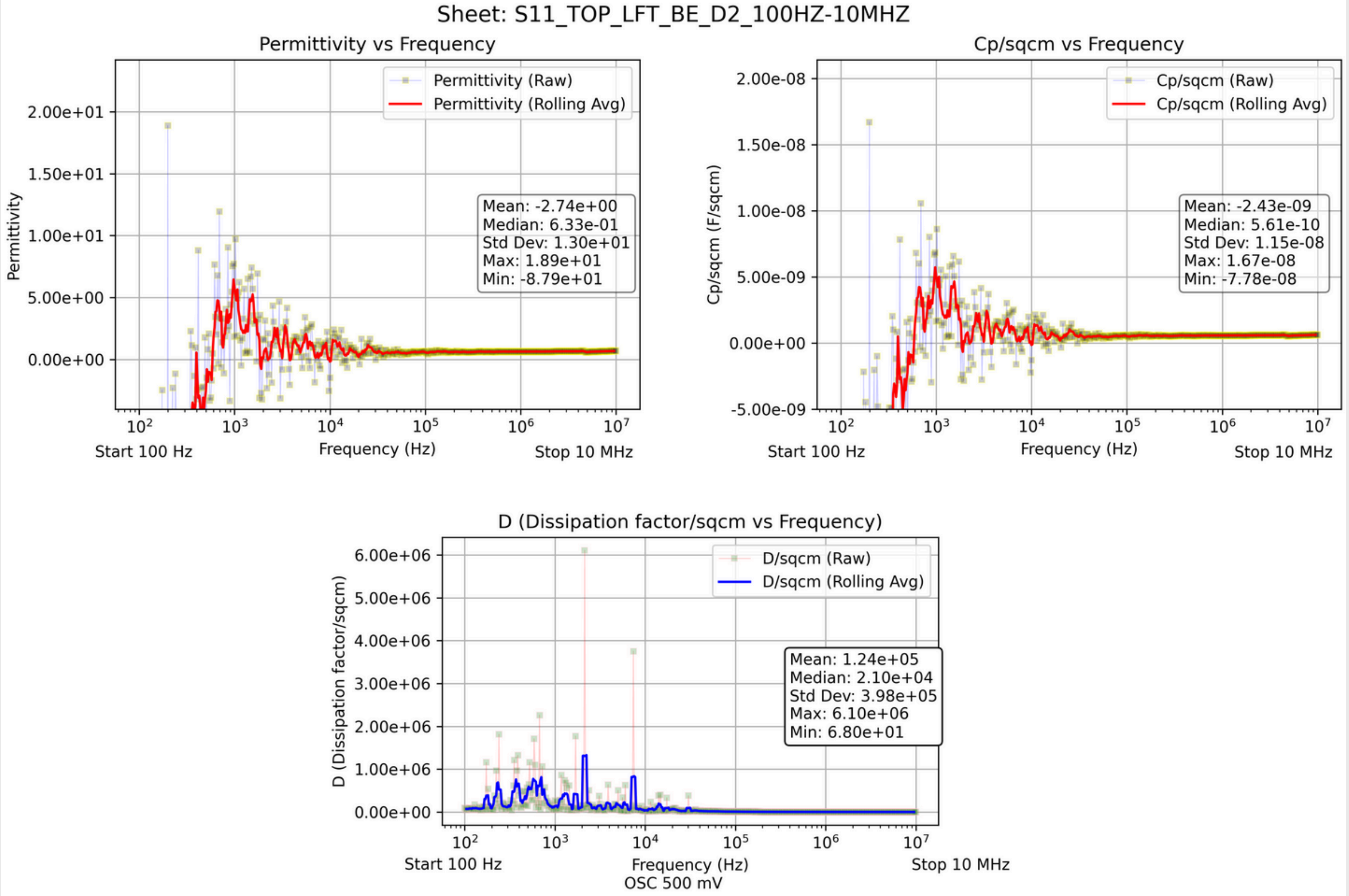


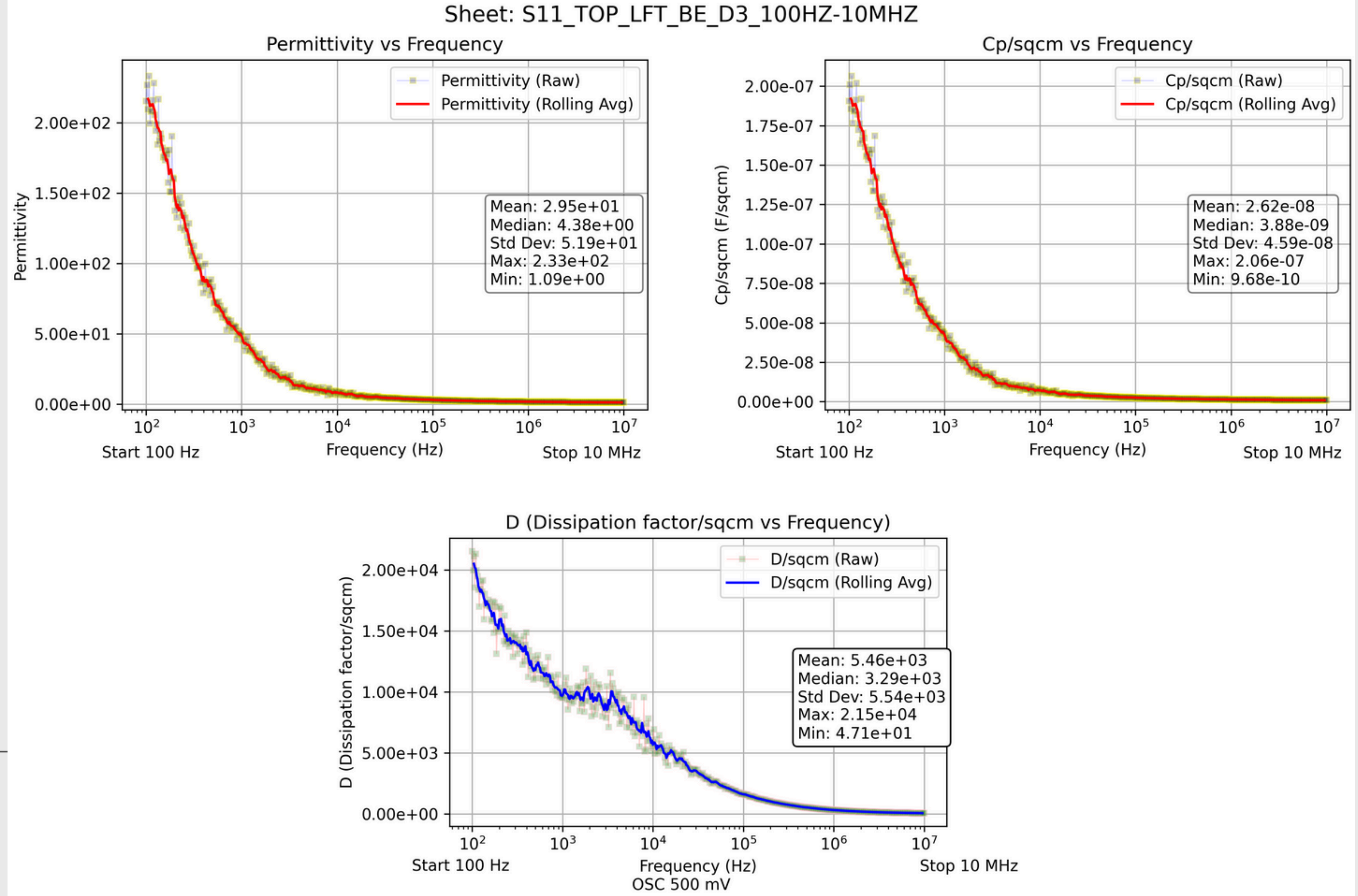
d33 calculation: https://drive.google.com/file/d/12Cyn9Ga3Ccr-Ry6bjPELuZiwxz8ZkNZC/view?usp=sharing

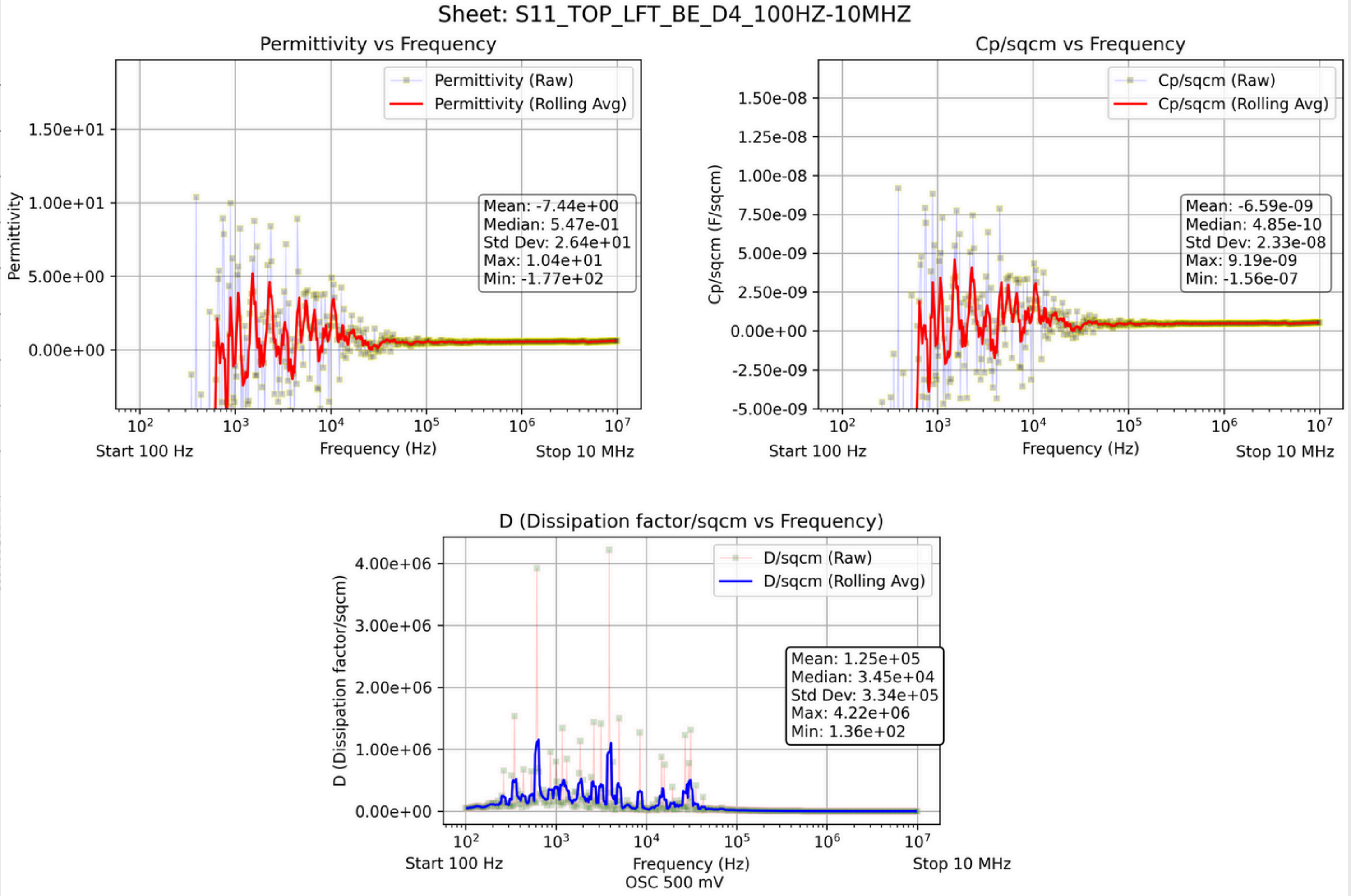
THANK YOU

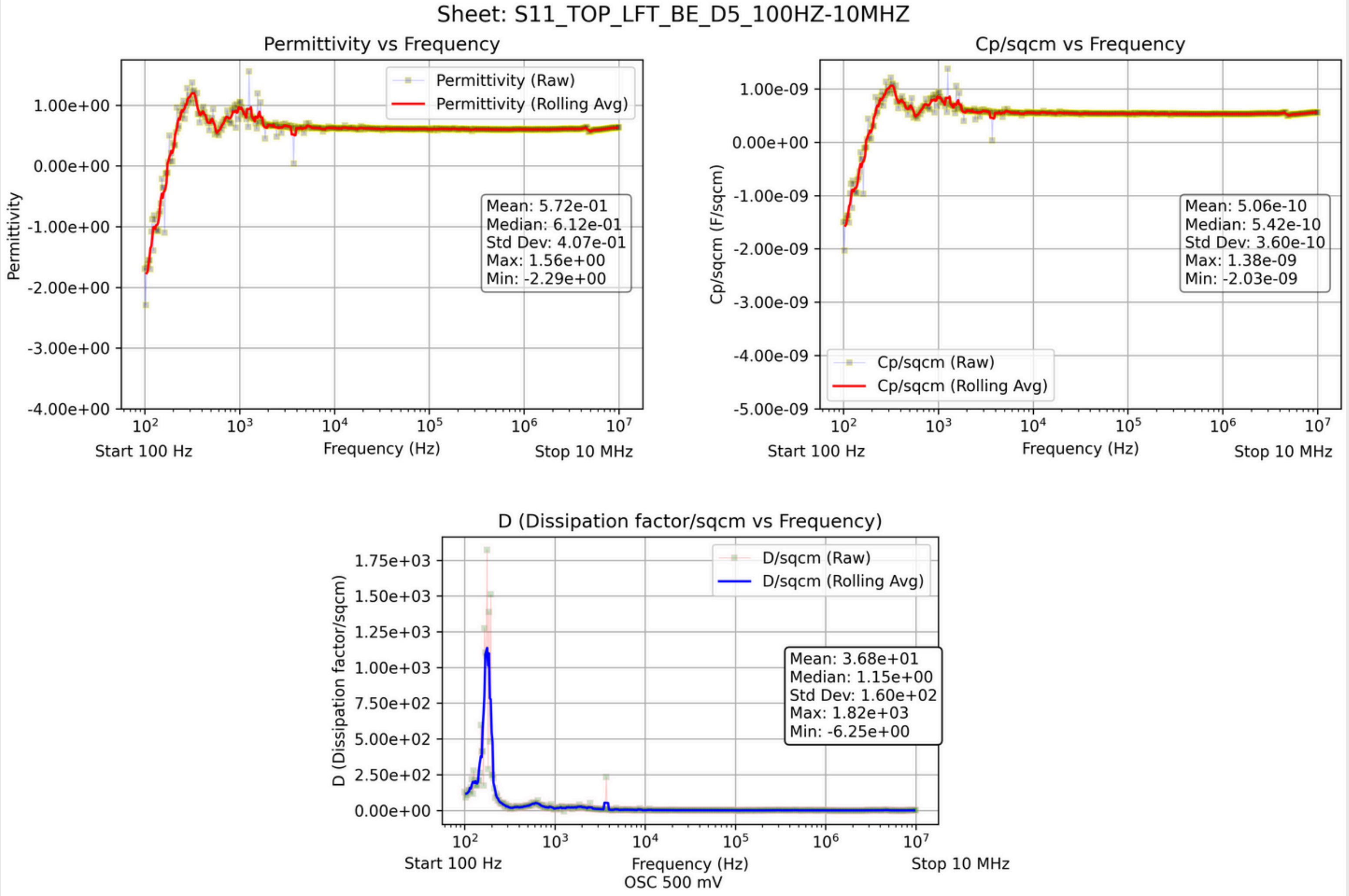
output: Device S11

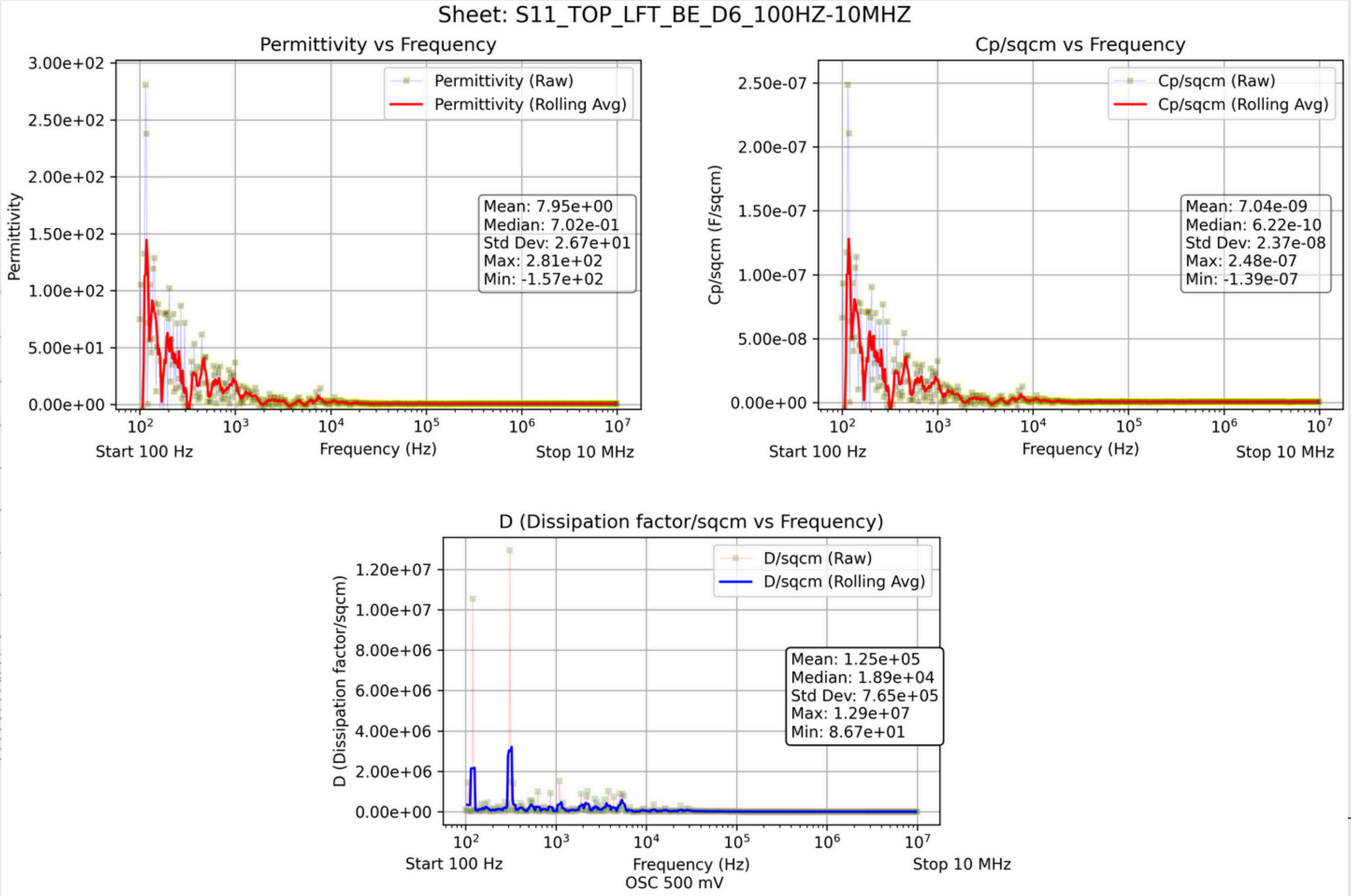


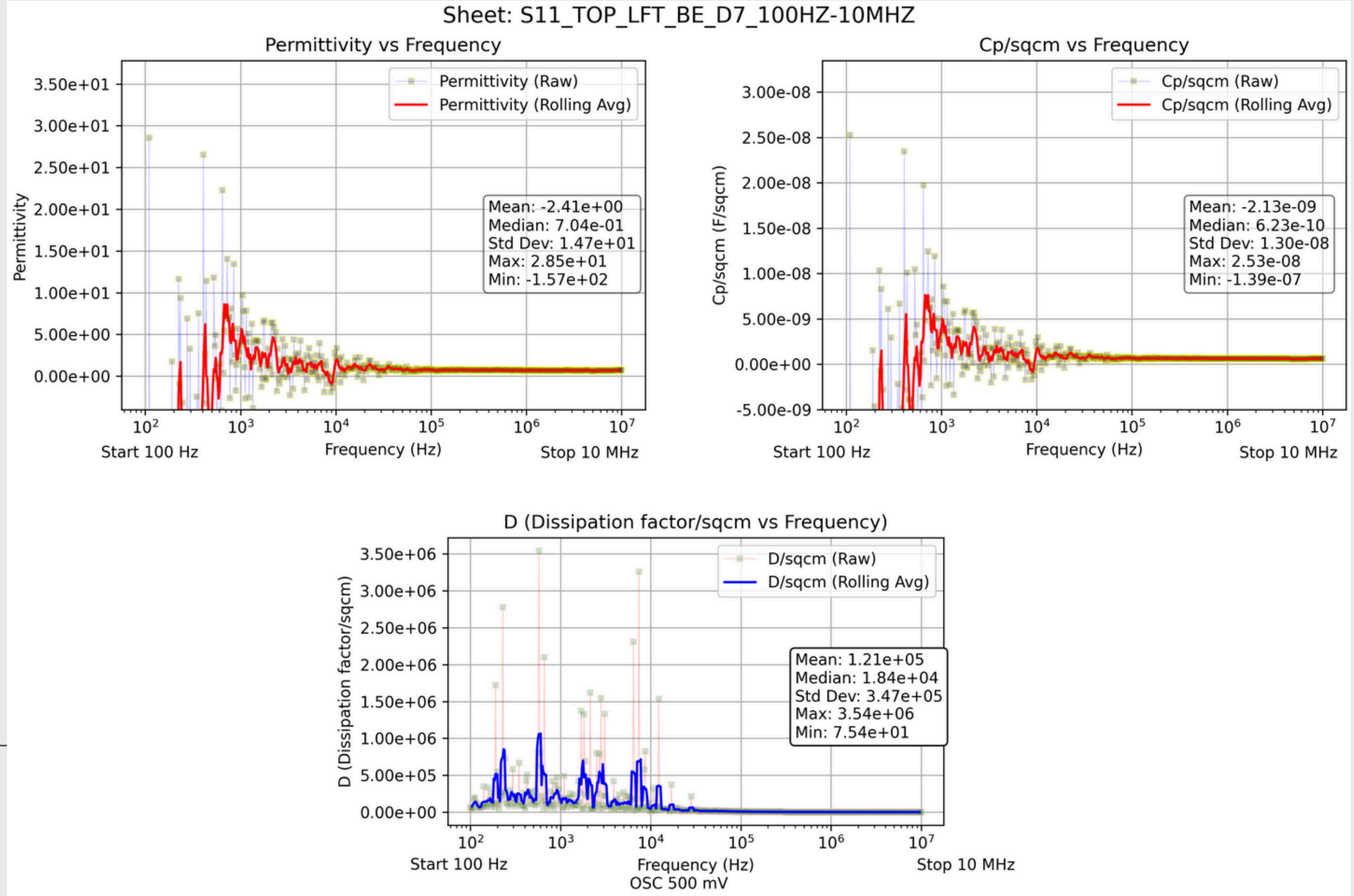





Sheet: S11_TOP_LFT_BE_D4_100HZ-10MHZ
Permittivity vs Frequency
Permittivity (Raw)
Permittivity (Rolling Avg)
Mean: -7.44e+00
Median: 5.47e-01
Std Dev: 2.64e+01
Max: 1.04e+01
Min: -1.77e+02
Permittivity
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
Cp/sqcm vs Frequency
Cp/sqcm (Raw)
Cp/sqcm (Rolling Avg)
Mean: -6.59e-09
Median: 4.85e-10
Std Dev: 2.33e-08
Max: 9.19e-09
Min: -1.56e-07
Cp/sqcm (F/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
D (Dissipation factor/sqcm vs Frequency)
D/sqcm (Raw)
D/sqcm (Rolling Avg)
Mean: 1.25e+05
Median: 3.45e+04
Std Dev: 3.34e+05
Max: 4.22e+06
Min: 1.36e+02
D (Dissipation factor/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
OSC 500 mV


Sheet: S11_TOP_LFT_BE_D5_100HZ-10MHZ
Permittivity vs Frequency
Permittivity (Raw)
Permittivity (Rolling Avg)
Mean: 5.72e-01
Median: 6.12e-01
Std Dev: 4.07e-01
Max: 1.56e+00
Min: -2.29e+00
Permittivity
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
Cp/sqcm vs Frequency
Cp/sqcm (Raw)
Cp/sqcm (Rolling Avg)
Mean: 5.06e-10
Median: 5.42e-10
Std Dev: 3.60e-10
Max: 1.38e-09
Min: -2.03e-09
Cp/sqcm (F/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
D (Dissipation factor/sqcm vs Frequency)
D/sqcm (Raw)
D/sqcm (Rolling Avg)
Mean: 3.68e+01
Median: 1.15e+00
Std Dev: 1.60e+02
Max: 1.82e+03
Min: -6.25e+00
D (Dissipation factor/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
OSC 500 mV


Sheet: S11_TOP_LFT_BE_D6_100HZ-10MHZ
Permittivity vs Frequency
Permittivity (Raw)
Permittivity (Rolling Avg)
Mean: 7.95e+00
Median: 7.02e-01
Std Dev: 2.67e+01
Max: 2.81e+02
Min: -1.57e+02
Permittivity
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
Cp/sqcm vs Frequency
Cp/sqcm (Raw)
Cp/sqcm (Rolling Avg)
Mean: 7.04e-09
Median: 6.22e-10
Std Dev: 2.37e-08
Max: 2.48e-07
Min: -1.39e-07
Cp/sqcm (F/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
D (Dissipation factor/sqcm vs Frequency)
D/sqcm (Raw)
D/sqcm (Rolling Avg)
Mean: 1.25e+05
Median: 1.89e+04
Std Dev: 7.65e+05
Max: 1.29e+07
Min: 8.67e+01
D (Dissipation factor/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
OSC 500 mV


Sheet: S11_TOP_LFT_BE_D7_100HZ-10MHZ
Permittivity vs Frequency
Permittivity (Raw)
Permittivity (Rolling Avg)
Mean: -2.41e+00
Median: 7.04e-01
Std Dev: 1.47e+01
Max: 2.85e+01
Min: -1.57e+02
Permittivity
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
Cp/sqcm vs Frequency
Cp/sqcm (Raw)
Cp/sqcm (Rolling Avg)
Mean: -2.13e-09
Median: 6.23e-10
Std Dev: 1.30e-08
Max: 2.53e-08
Min: -1.39e-07
Cp/sqcm (F/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
D (Dissipation factor/sqcm vs Frequency)
D/sqcm (Raw)
D/sqcm (Rolling Avg)
Mean: 1.21e+05
Median: 1.84e+04
Std Dev: 3.47e+05
Max: 3.54e+06
Min: 7.54e+01
D (Dissipation factor/sqcm)
Start 100 Hz
Frequency (Hz)
Stop 10 MHz
OSC 500 mV